

6 Frame Replication and Elimination for Reliability

8 **Corrigendum 1: Technical and Editorial Corrections**

11 **LAN** **MAN** **Standards** **Committee**
12 of the
13 **IEEE Computer Society**

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Piscataway, NJ 08854, USA

1 **Abstract:** This corrigendum to IEEE Std 802.1CB™-2017 provides technical and editorial
2 corrections.

3 **Keywords:** TSN, Time-Sensitive Networking, Redundancy, Bridging, Bridges, Frame Replication,
4 Frame Elimination, Stream identification, Extended Stream identification, Bridged Local Area
5 Networks, IEEE 802®, IEEE 802.1Q™, IEEE 802.1CB™, IEEE 802.1CBdb, local area networks
6 (LANs), MAC Bridges, Virtual Bridged Local Area Networks (virtual LANs), amendment.

1 Editor's Foreword

2 <<Notes>>

3 <<Throughout this document, all notes such as this one, presented between angle braces, are temporary
4 notes inserted by the Editors for a variety of purposes; these notes and the Editors' Foreword will all be
5 removed prior to publication and are not part of the normative text.>>

6 <<Comments and participation in 802.1 standards development

7 Comments on this draft are encouraged. **PLEASE NOTE: All issues related to IEEE standards
8 presentation style, formatting, spelling, etc. are routinely handled between the 802.1 Editor and the
9 IEEE Staff Editors prior to publication, after balloting and the process of achieving agreement on the
10 technical content of the standard is complete.** Readers are urged to devote their valuable time and energy
11 only to comments that materially affect either the technical content of the document or the clarity of that
12 technical content. Comments should not simply state what is wrong, but also what might be done to fix the
13 problem.

14 Full participation in the development of this draft requires individual attendance at IEEE 802 meetings.
15 Information on 802.1 activities, working papers, and email distribution lists etc. can be found on the 802.1
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17 <https://1.ieee802.org/>

18 Use of the email distribution list is not presently restricted to 802.1 members, and the working group has had
19 a policy of considering ballot comments from all who are interested and willing to contribute to the
20 development of the draft. Individuals not attending meetings have helped to identify sources of
21 misunderstanding and ambiguity in past projects. Non-members are advised that the email lists exist primarily
22 to allow the members of the working group to develop standards, and are not a general forum.

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24 familiarize themselves with that policy. See <http://standards.ieee.org/about/sasb/patcom/materials.html>.

25 Comments on this document may be sent to the 802.1 email exploder, to the editors, or to the Chairs of the
26 802.1 Working Group and Time-Sensitive Networking Task Group.

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32 Email: janos.farkas@ericsson.com Email: glenn.parsons@ericsson.com

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34 may not be acknowledged. >>**

35 <<The draft text and accompanying information

36 This document currently comprises:

- 37 — A title page for the proposed standard including an Abstract and Keywords. This title page will be retained following working group approval of this draft, i.e. prior to sponsor ballot.

- 1 — The editors' forewords, including this text. These include an unofficial and informal appraisal of history and status, introductory notes to each draft that summarize the progress and focus of each successive draft, and requests for comments and contributions on major issues.
- 4 — IEEE boilerplate text.
- 5 — A record of participants (not included in early drafts but added prior to publication).
- 6 — The introduction to this standard.
- 7 — The proposed standard proper.

8 During the early stages of draft development, 802.1 editors have a responsibility to attempt to craft technically
9 coherent drafts from the resolutions of ballot comments and the other discussions that take place in the
10 working group meetings. Preparation of drafts often exposes inconsistencies in editors instructions or
11 exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices
12 and requests by the editors' for contributions on specific issues will be found in the editors' introductory notes
13 to the current draft, at appropriate points in the draft, and in Annex Z. Significant discussion of more difficult
14 topics will be found in the last of these.

15 The ballot comments received on each draft, and the editors' proposed and final disposition of comments, are
16 part of the audit trail of the development of the standard and are available, along with all the revisions of the
17 draft on the 802.1 web site (for address see above).

18 >>

19 << **Introductory notes to draft 1.0**

20 Draft 1.0 was prepared for a first Working Group ballot, as a result of comment resolution on Draft 0.1, carried
21 out at the IEEE 802.1 March 2025 virtual meeting. The ballot comment resolution is documented in 802-1CB-
22 2017-cor1-d0-1-dis-v01.pdf. Changes from Draft 0.1 include:

- 23 — various typo corrections;
- 24 — Consistency between modified figures and corresponding descriptive text;
- 25 — Alignment of the Stream_handle value definition between clauses 6.2 and 9.1.1.1.

26 >>

27 << **Project Authorization Request, Scope, Purpose, and Five Criteria**

28 A PAR (Project Authorization Request) for P802.1CB-2017/Cor1 was approved by the IEEE Standards
29 Association on February 15, 2024. The following information is taken from the 802.1CB-2017/Cor1 PAR.

30 **Scope of Proposed Changes:**

31 Correction of technical and editorial errors identified and agreed by the IEEE 802.1 Working Group
32 maintenance activity comprising: <https://www.802-1.org/items/431>, <https://www.802-1.org/items/451>, <https://www.802-1.org/items/456>, and <https://www.802-1.org/items/471>. The management module changes needed
34 are applicable in the context of IEEE Std 802.1CBv(TM)-2021..

35 **Purpose of Proposed Project:**

36 This document will not include a purpose clause.

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1 Participants

2 The following is a list of participants in the IEEE 802.1 Working Group.

3 **Glenn Parsons, *Chair***

4 **Jessy Rouyer, *Vice-Chair***

5 **Mark Hantel, *Chair, Maintenance Task Group***

6 **Christophe Mangin, *Editor***

7

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3

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5
6 Name, *DOE Representative*
7 Name, *NIST Representative*
8
9
10 Name
11 IEEE Standards Program Manager, Document Development
12
13
14 Name
15 IEEE Standards Program Manager, Technical Program Development

1 Introduction

2

This introduction is not part of IEEE P 802.1CB-2017/Cor1-20XX, IEEE Draft Standard for Local and metropolitan area networks – Frame Replication and Elimination for Reliability – Corrigendum 1.

3 The first edition of IEEE Std 802.1CB was published in 2017. A first amendment IEEE Std 802.1CBv-
4 2021 added the YANG data model that allows configuration and state reporting for all managed objects of
5 the base standard. A second amendment IEEE Std 802.1CBdb-2021 added the mask-and-match Stream
6 identification function and provided technical and editorial corrections.

7 This corrigendum IEEE 802.1CB-2017/Cor1-2025 implements the changes necessitated by various mainte-
8 nance items that have been notified to the IEEE 802.1 Working Group. The maintenance items addressed by
9 this corrigendum are: .315, 338, 343, 344 and 364.

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1

2 Draft Standard for 3 Local and metropolitan area networks — 4 Frame Replication and Elimination for 5 Reliability 6 Corrigendum 1: Technical end Editorial 7 Corrections

8 NOTE—The editing instructions contained in this amendment define how to merge the material contained here into the
9 base document and its other amendments to form the new comprehensive standard.

10 The editing instructions are shown in ***bold italic***. Four editing instructions are used: ***change***, ***delete***, ***insert***, and ***replace***.
11 ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change
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14 may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make
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17 incorporated into the base standard.⁶

⁶Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

1 **6. Stream identification**

2 **6.2 Stream identification function**

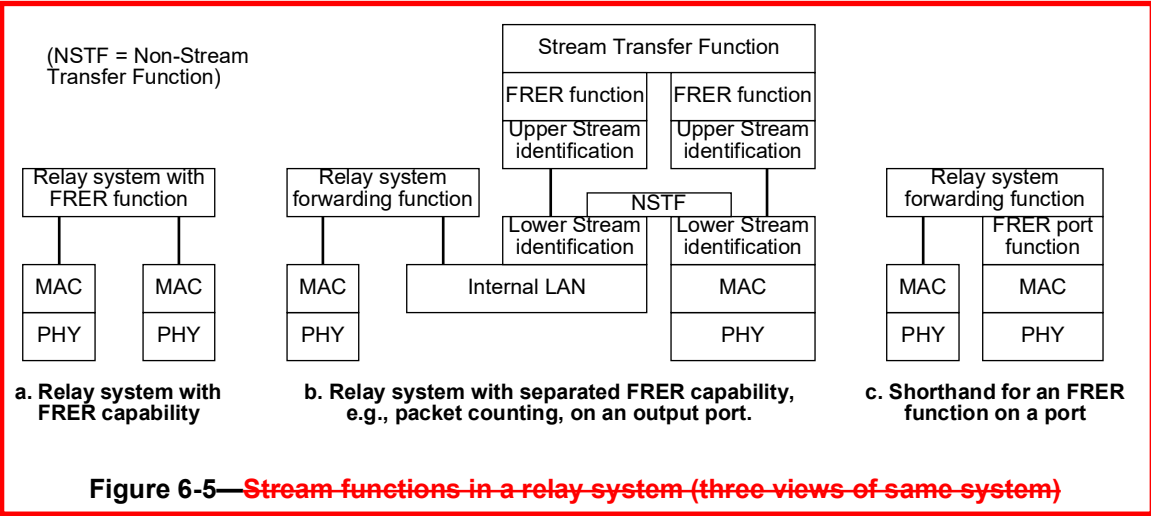
3 *Insert the following paragraph at the end of the clause, before the notes, in IEEE Std 802.1CB-2017:*

5 The null-value is a value known from the management system as identifying the stream_handle assigned to
6 packets that belong to no known Stream.

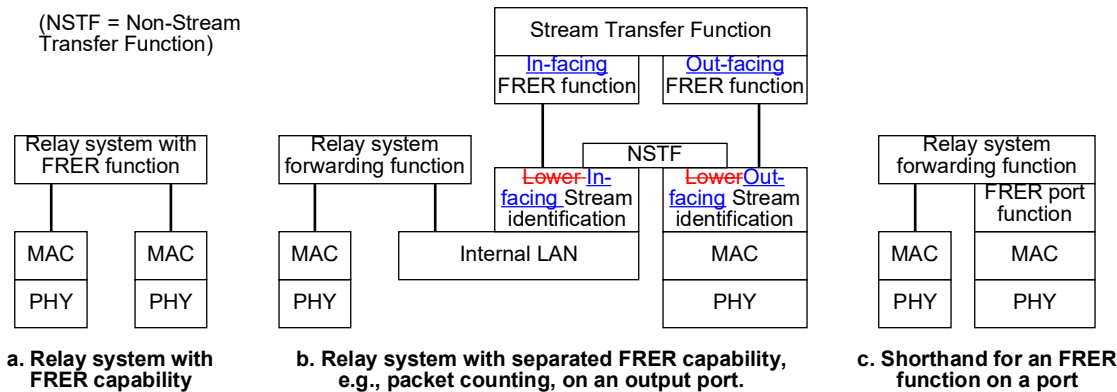
7 **6.3 Stream identification in systems**

8 *Change Figure 6-5 as shown:*

9 .



10 .



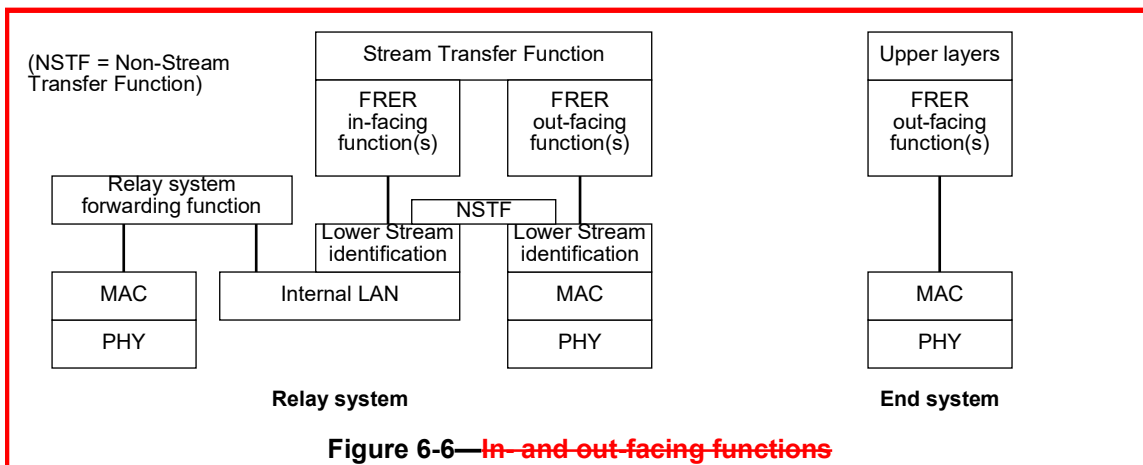
11

1 *Change the text of the second paragraph as shown:*

2 Diagram b in Figure 6-5 shows that same relay system, this time with the FRER functions placed explicitly
3 in one of its ports, the picture of which is greatly expanded. In Diagram b, the forwarding function has no
4 FRER capabilities. Stream identification functions extract Stream identification function subparameters in
5 order to enable an FRER function. The connectivity between the Relay system's forwarding and the Stream
6 identification functions is modelled by an Internal LAN that provides transparent connectivity between ISS,
7 or EISS, SAPs supporting the Stream service subparameters (6.1). The *Stream Transfer Function* acts as a
8 two-port packet relay, residing entirely inside the one port of the relay system, relaying packets belonging to
9 Streams. The *Non-Stream Transfer Function* (NSTF) does the same, but attaches to the “unknown” SAP of
10 Figure 6-4, and thus relays packets not recognized as belonging to Streams. The *Lower In-facing and Out-*
11 *facing Stream identification* functions separate FRER packets from non-FRER packets; the latter are relayed
12 across the NSTF as if the FRER capabilities were not present. ~~The Upper Stream identification functions~~
13 ~~identify the FRER packets' Streams so that the other FRER functions can perform their tasks.~~ Each of the
14 two-port transfer functions have a pair of SAPs, and transfer all packet receive indications into packet
15 requests on the other SAP. This formulation illustrates exactly the peering relationships among the various
16 functions, but is too complex for many purposes.

17 *Change Figure 6-6 as shown:*

18



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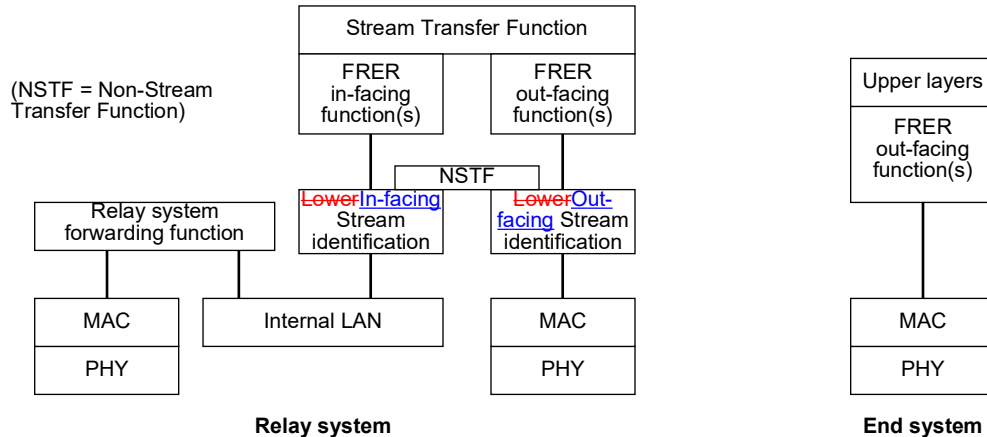


Figure 6-6—In- and out-facing functions

6.4 Null Stream identification

Change the text of the third sentence of the first paragraph as shown:

~~It discards the stream_handle subparameter passed down the stack.~~

6.5 Source MAC and VLAN Stream identification

Change the text of the third sentence of the first paragraph as shown:

~~It discards the stream_handle subparameter passed down the stack.~~

6.7 IP Stream identification

Change the text of the third sentence of the first paragraph as shown:

In IP Stream identification, the IP and higher layer address parameters and the EISS parameters are used to determine the stream_handle subparameter of packets passed up the stack, ~~and discards the stream_handle subparameter for packets passed down the stack.~~

Change the text of the second paragraph as shown:

IP Stream identification can be coupled, for example, with Active Destination MAC and VLAN Stream identification (6.6) to assign a particular {MAC address, VLAN, priority} triple to packets belonging to a particular unicast IP Stream, as shown in Figure 8-1, Port A, where IP Stream identification would be in the box labeled “Passive ~~Upper~~ Out-facing Stream identification functions (6-26.3).”

Insert the following text after the second paragraph:

Practically, the IP Stream identification function requires to locate the beginning of the IP packet encapsulated in the frame, preceded by an IP Ethertype field, e.g. 0x0800 or 0x88DD. The location of the IP Ethertype field within the frame is dependent on the combination of Tag(s) following the addresses in a frame. The method used to determine this location is beyond the scope of this standard.

1 6.8 Mask-and-match Stream identification

2 *Change the text of the second paragraph as shown:*

3 ~~It discards the stream_handle subparameter passed down the stack.~~ It generates a stream_handle
4 subparameter on frames passed up the stack based on information extracted from a subset of the frame's ISS
5 parameters. It does not change any of a packet's parameters, other than the connection_identifier.

6 *Change the text of the last paragraph as shown:*

7 Mask-and-match Stream identification can be coupled, for example, with Active Destination MAC and
8 VLAN Stream identification (6.6) to assign a particular {MAC address, VLAN, priority} triplet to packets
9 belonging to a particular unicast application Stream identified by the combination of the values of particular
10 fields in the MAC service data unit, as shown in Figure 8-1, Port A, where Mask-and-match Stream
11 identification would be in the box labeled "Passive ~~Upper~~Out-facing Stream identification functions
12 (~~6.26.3~~)."

8. Frame Replication and Elimination for Reliability in Bridges

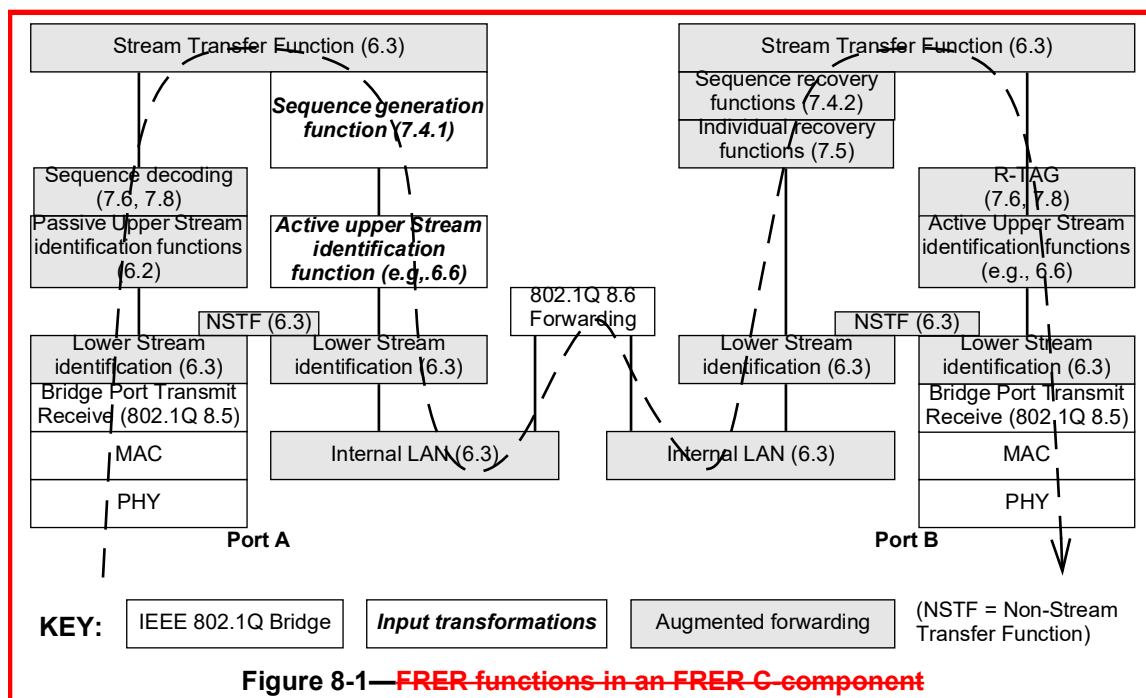
8.1 Limiting options

Change the text of the second paragraph as shown:

Most of the flexibility of the model described in Clause 6 and Clause 7 can be achieved in an IEEE 802.1Q Bridge, if we limit the Bridge to two stages of activity:

- 1) **Input transformations:** An expanded port (as in, e.g., Figure C-5) that includes Sequence generation functions (7.4.1) and active **upper-in-facing** Stream identification functions (6.6), shown in white boxes with boldface type in Figure 8-1.
- 2) **Augmented forwarding:** The Bridge Forwarding Process, described in 8.6 of IEEE Std 802.1Q-2014 (shown in white boxes in Figure 8-1 [and Figure 8-2](#)), augmented with Stream identification functions (6.2, both incoming and outgoing), Sequence encode/decode functions (7.6, both incoming and outgoing), Sequence recovery functions (7.4.2), Individual recovery functions (7.5), and their associated infrastructure, all shown in shaded boxes in Figure 8-1 [and Figure 8-2](#).

Replace Figure 8-1 by Figures 8-1 and 8-2 as shown and renumber the subsequent figures accordingly:



1

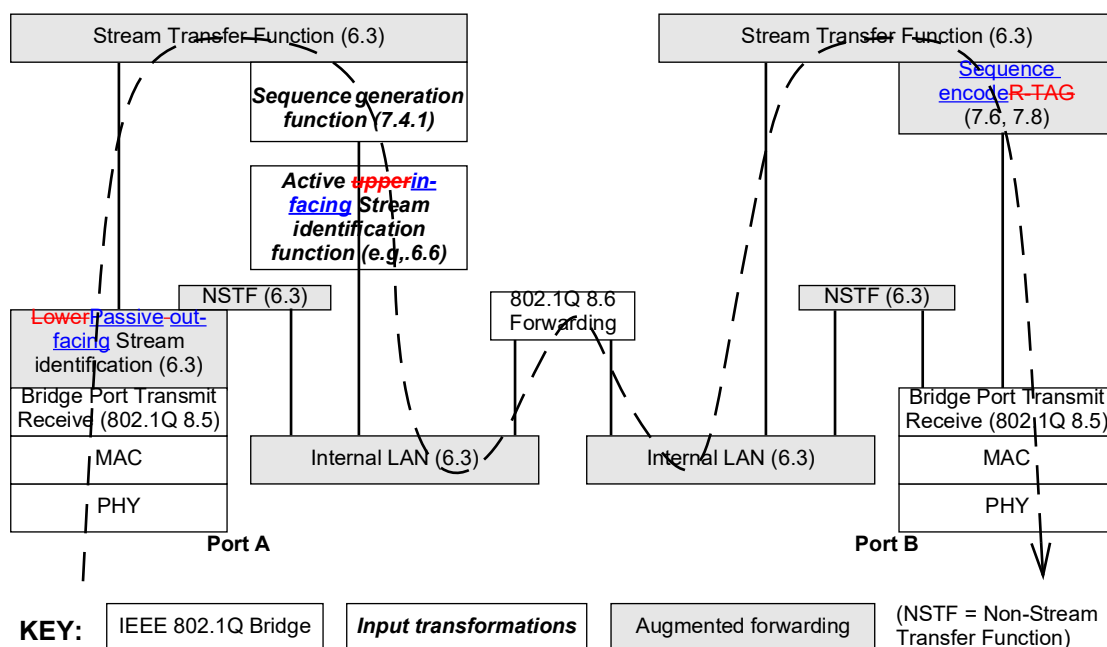


Figure 8-1—Split functions in an FRER C-component

2

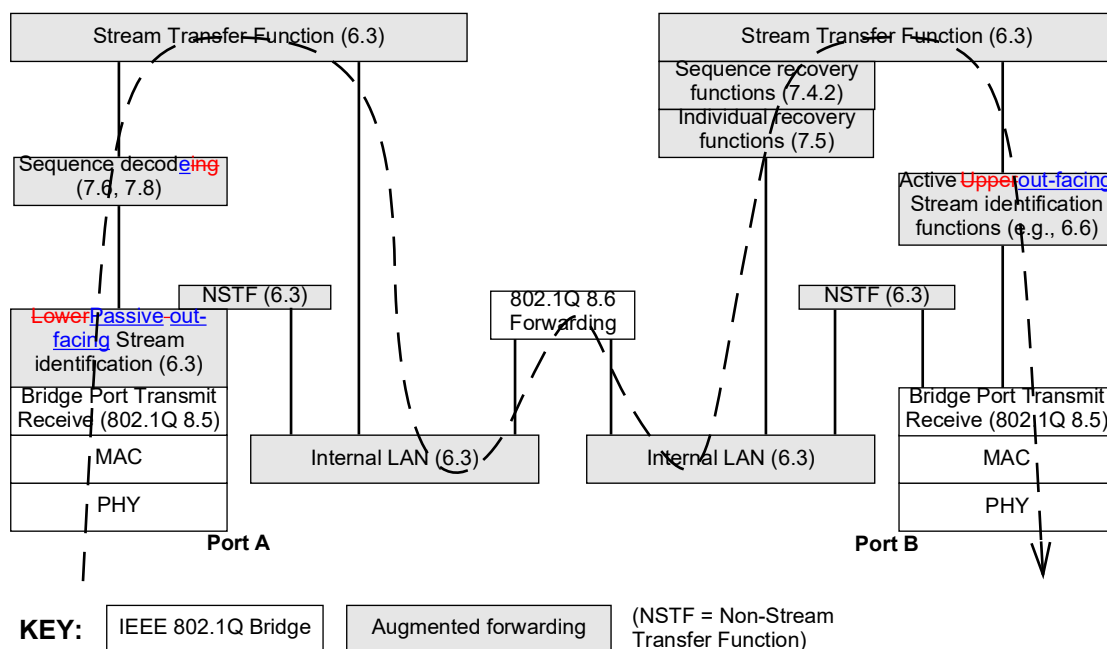


Figure 8-2—Merge functions in an FRER C-component

3 *Change the text of the fifth and sixth paragraphs as shown:*

4 As shown in [Figure 8-1](#)[Figure 8-2](#), the paradigm is that a given Stream is recognized on input, meaning that
5 the stream_handle and sequence_number subparameters are extracted. The frame is then forwarded
6 normally through the Bridge. On the output port, the stream-handle and sequence_number subparameters

1 found on the input port are used to drive the Individual recovery functions, Sequence recovery functions,
2 Sequence encoding, and Stream identification functions.

3 Figure 8-1 ~~and Figure 8-2~~ ~~does~~ not imply that the Augmented forwarding functions have to be placed in
4 expanded ports; they can be integrated with the Bridge forwarding function (8.6 of IEEE 802.1Q-2014) as
5 shown in ~~Figure 8-2~~ [Figure 8-3](#). The ordering of this functions with respect to 8.6 of IEEE 802.1Q-2014 can be
6 important. In particular:

7 **8.3 Frame Replication and Elimination for Reliability and VLAN tags**

8 *Change the text as shown:*

9 As illustrated in Figure 8-1, [Figure 8-2](#) and Figure 8-3, FRER in an IEEE 802.1Q C-VLAN Component is
10 above the Bridge Port Transmit and Receive function (8.5 of IEEE Std 802.1Q-2014) in the protocol stack.
11 As a consequence of this placement, a frame containing both an IEEE 802.1Q C-VLAN tag and an R-TAG
12 but no other IEEE 802.1Q tags, would be as illustrated in ~~Figure 8-3~~ [Figure 8-4](#)

13 **8.4 Configuring Frame Replication and Elimination for Reliability in Bridges**

14 *Change the text as shown:*

15 As described in 8.1, ~~we can describe~~ the output transformations, including Individual recovery functions
16 (7.5), Sequence recovery functions (7.4.2), Sequence encode/decode functions (7.6), and Stream
17 identification functions (6.2), as being embedded in the IEEE 802.1Q Forwarding Process. However, since
18 different activities can be required by an application on different output ports, the managed objects in
19 Clause 9 and Clause 10 can be used to configure an FRER C-component. Figure 8-1 [and Figure 8-2](#)
20 illustrates an allowed arrangement of FRER functions on an output Bridge port as perceived by the managed
21 objects in Clause 9 and Clause 10.

22

9. Stream Identification Management

9.1 Stream identity table

9.1.1 tsnStreamIdEntry

9.1.1.1 tsnStreamIdHandle

Insert the following note at the end of the subclause:

NOTE—It is also possible to assign a packet that was identified as belonging to a known stream (i.e., a specific tsn-StreamIdEntry) a stream handle equal 0.

8

1 12. YANG Data Model

2 12.1 YANG Framework

3 *Change the text in the lettered-list as shown:*

4 This clause has been developed according to the YANG guidelines published in IETF RFC 6087 [B5] as
5 applicable to IEEE standards.

6 The YANG framework applies hierarchy in the following areas:

- 7 a) The uniform resource name (URN), as specified in IEEE Std 802d [B2a]. ~~The structure of the URN~~
8 ~~is such that ieee is the root (i.e., name-space identifier), followed by the standard, then the working~~
9 ~~group developing the standard.~~
- 10 b) The YANG objects form a hierarchy of configuration and operational data structures that define the
11 YANG model. These hierarchical relationships are described in 12.2.

12 12.2 IEEE Std 802.1CB YANG model

13 12.2.2 Frame Replication and Elimination for Reliability model

14 *Replace Figure 12-5 with the following figure:*

1

sequence-generation (10.3)		
uint32	index	// r-w
handle-ref	* stream	// r-w
boolean	direction-out-facing	// r-w
boolean	reset	// r-w

sequence-recovery (10.4)		
uint32	index	// r-w
handle-ref	* stream	// r-w
if-ref	* port	// r-w
boolean	direction-out-facing	// r-w
boolean	reset	// r-w
sequence-recovery-algorithm	algorithm	// r-w
sequence-history-length	history-length	// r-w
uint32	reset-timeout	// r-w
uint32	invalid-sequence-value	// r
boolean	take-no-sequence	// r-w
boolean	individual-recovery	// r-w
boolean	latent-error-detection	// r-w
int32	latent-error-difference	// r-w
uint32	latent-error-period	// r-w
uint16	latent-error-paths	// r-w
uint32	latent-error-reset-period	// r-w

sequence-identification (10.5)		
handle-ref	* stream	// r-w
if-ref	port	// r-w
boolean	direction-out-facing	// r-w
boolean	active	// r-w
enum	encapsulation	// r-w
int8	path-id-lan-id	// r-w

stream-split (10.6)		
uint32	index	// r-w
if-ref	port	// r-w
boolean	direction-out-facing	// r-w
leaf-ref	* input-id	// r-w
leaf-ref	* output-id	// r-w

sequence-autoconfiguration (10.7.1)		
uint32	index	// r-w
enum	sequence-encapsulation	// r-w
if-ref	* receive-port	// r-w
enum	tagged	// r-w
dot1qtypes:vlanid	* vlan	// r-w
if-ref	* recovery-port	// r-w
uint64	destruction-interval	// r-w
uint64	reset-interval	// r-w
sequence-recovery-algorithm	algorithm	// r-w
sequence-history-length	history-length	// r-w
boolean	create-individual	// r-w
boolean	create-recovery	// r-w
boolean	latent-error-detection	// r-w
int32	latent-error-difference	// r-w
uint32	latent-error-period	// r-w
uint32	latent-error-reset-period	// r-w

output-autoconfiguration (10.7.2)		
uint32	index	// r-w
if-ref	* port	// r-w
enum	encapsulation	// r-w
int8	lan-path-id	// r-w

Figure 12-5—FRER model

12.6 Definition of 802.1CB YANG modules^{7 8}

12.6.1.2 YANG data scheme definition for ieee802-dot1cb-frer YANG module

Change the text as shown:

```

4 module: ieee802-dot1cb-frer
5   +--rw frer
6     +--rw sequence-generation* [index]
7 |   +--rw index                               uint32
8     |   +--rw stream*                         -> /dot1cb-sid:stream-identity/handle
9     |   +--rw direction-out-facing?          dot1cb-sid-types:direction
10    |   +--rw reset?                          boolean
11    +--rw sequence-recovery* [index]
12    |   +--rw index                           uint32
13    |   +--rw stream*
14    |   |   -> /dot1cb-sid:stream-identity/handle
15    |   +--rw port*                          if:interface-ref
16    |   +--rw direction-out-facing?          dot1cb-sid-types:direction
17    |   +--rw reset?                          boolean
18    |   +--rw algorithm
19    |   |   +--rw (algorithm)?
20    |   |   |   +--:(vector)
21    |   |   |   |   +--rw vector!
22    |   |   |   |   +--ro type-number?      dot1cb-frer-types:seq-rcvy-algorithm
23    |   |   |   |   +--ro oui-cid?          string
24    |   |   |   +--:(match)
25    |   |   |   |   +--rw match!
26    |   |   |   |   +--ro type-number?      dot1cb-frer-types:seq-rcvy-algorithm
27    |   |   |   |   +--ro oui-cid?          string
28    |   |   |   +--:(organization-specific)
29    |   |   |   |   +--rw organization-specific!
30    |   |   |   |   +--rw type-number?      int32
31    |   |   |   |   +--rw oui-cid?          string
32    |   +--rw history-length?                 sequence-history-length
33    |   +--rw reset-timeout?                  uint32
34    |   +--ro invalid-sequence-value?         uint32
35    |   +--rw take-no-sequence?               boolean
36    |   +--rw individual-recovery?            boolean
37    |   +--rw latent-error-detection?         boolean
38    |   +--rw latent-error-detection-parameters
39    |   |   +--rw difference?                  int32
40    |   |   +--rw period?                     uint32
41    |   |   +--rw paths?                      uint16
42    |   |   +--rw reset-period?               uint32
43    +--rw sequence-identification* [port direction-out-facing]
44    |   +--rw stream*                         -> /dot1cb-sid:stream-identity/handle
45    |   +--rw port                           if:interface-ref
46    |   +--rw direction-out-facing          dot1cb-sid-types:direction
47    |   +--rw active?                       boolean
48    |   +--rw encapsulation
49    |   |   +--rw (encapsulation)?
50    |   |   |   +--:(r-tag)

```

⁷Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

⁸An ASCII version of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

```

1 | | | +--rw r-tag!
2 | | | +--ro type-number? dot1cb-frer-types:seq-encaps-method
3 | | | +--ro oui-cid? string
4 | | | +--:(hsr-sequence-tag)
5 | | | | +--rw hsr-sequence-tag!
6 | | | | +--ro type-number? dot1cb-frer-types:seq-encaps-method
7 | | | | +--ro oui-cid? string
8 | | | +--:(prp-sequence-tag)
9 | | | | +--rw prp-sequence-tag!
10 | | | | +--ro type-number? dot1cb-frer-types:seq-encaps-method
11 | | | | +--ro oui-cid? string
12 | | | +--:(organization-specific)
13 | | | +--rw organization-specific!
14 | | | +--rw type-number? int32
15 | | | +--rw oui-cid? string
16 | | +--rw path-id-lan-id? lan-path-id
17 +--rw stream-split* [port direction-out-facing index]
18 | +--rw index uint32
19 | +--rw port if:interface-ref
20 | +--rw direction-out-facing dot1cb-sid-types:direction
21 | +--rw input-id* -> /dot1cb-sid:stream-identity/handle
22 | +--rw output-id* -> /dot1cb-sid:stream-identity/handle
23 +--rw autoconfiguration {auto-configuration}?
24 +--rw sequence* [index]
25 | +--rw index uint32
26 | +--rw sequence-encapsulation
27 | | +--rw (encapsulation)?
28 | | | +--:(r-tag)
29 | | | | +--rw r-tag!
30 | | | | +--ro type-number?
31 | | | | | dot1cb-frer-types:seq-encaps-method
32 | | | | +--ro oui-cid? string
33 | | | +--:(hsr-sequence-tag)
34 | | | | +--rw hsr-sequence-tag!
35 | | | | +--ro type-number?
36 | | | | | dot1cb-frer-types:seq-encaps-method
37 | | | | +--ro oui-cid? string
38 | | | +--:(prp-sequence-tag)
39 | | | | +--rw prp-sequence-tag!
40 | | | | +--ro type-number?
41 | | | | | dot1cb-frer-types:seq-encaps-method
42 | | | | +--ro oui-cid? string
43 | | | +--:(organization-specific)
44 | | | +--rw organization-specific!
45 | | | +--rw type-number? int32
46 | | | +--rw oui-cid? string
47 | +--rw receive-port* if:interface-ref
48 | +--rw tagged? enumeration
49 | +--rw vlan* dot1qtypes:vlanid
50 | +--rw recovery-port* if:interface-ref
51 | +--rw destruction-interval? uint64
52 | +--rw reset-interval? uint64
53 | +--rw algorithm
54 | | +--rw (algorithm)?
55 | | | +--:(vector)
56 | | | | +--rw vector!
57 | | | | +--ro type-number?
58 | | | | | dot1cb-frer-types:seq-rcvy-algorithm
59 | | | | +--ro oui-cid? string

```

```

1      | |      +---: (match)
2      | |      | +---rw match!
3      | |      | +---ro type-number?
4      | |      | |      dotlcb-frer-types:seq-rcvy-algorithm
5      | |      | +---ro oui-cid?      string
6      | |      +---: (organization-specific)
7      | |      +---rw organization-specific!
8      | |      +---rw type-number?    int32
9      | |      +---rw oui-cid?      string
10     | +---rw history-length?      sequence-history-length
11     | +---rw create-individual?    boolean
12     | +---rw create-recovery?      boolean
13     | +---rw latent-error-detection? boolean
14     | +---rw latent-error-difference? int32
15     | +---rw latent-error-period?   uint32
16     | +---rw latent-error-reset-period? uint32
17     +---rw output* [index]
18         +---rw index                uint32
19         +---rw port*                if:interface-ref
20         +---rw encapsulation
21         | +---rw (encapsulation)?
22         | +---: (r-tag)
23         | | +---rw r-tag!
24         | | +---ro type-number?
25         | | |      dotlcb-frer-types:seq-encaps-method
26         | | +---ro oui-cid?      string
27         | +---: (hsr-sequence-tag)
28         | | +---rw hsr-sequence-tag!
29         | | +---ro type-number?
30         | | |      dotlcb-frer-types:seq-encaps-method
31         | | +---ro oui-cid?      string
32         | +---: (prp-sequence-tag)
33         | | +---rw prp-sequence-tag!
34         | | +---ro type-number?
35         | | |      dotlcb-frer-types:seq-encaps-method
36         | | +---ro oui-cid?      string
37         | +---: (organization-specific)
38         | +---rw organization-specific!
39         | +---rw type-number?    int32
40         | +---rw oui-cid?      string
41         +---rw lan-path-id?      lan-path-id
42
43     augment /dotlcb-sid:stream-identity:
44         +---ro auto-configured?    boolean {auto-configuration}?
45         +---rw lan-path-id?        lan-path-id {auto-configuration}?
46     augment /if:interfaces/if:interface/if:statistics:
47         +---ro frer
48         +---ro per-port-counters
49         | +---ro rx-passed-pkts?    uint64
50         | +---ro rx-discarded-pkts? uint64
51         | +---ro encode-errored-pkts? uint64
52         +---ro per-port-per-stream-counters* [direction-out-facing handle]
53         +---ro direction-out-facing dotlcb-sid-types:direction
54         +---ro handle
55         | -> /dotlcb-sid:stream-identity/handle
56         +---ro generation-reset?    uint64
57         +---ro rx-out-of-order-pkts? uint64
58         +---ro rx-roguer-pkts?      uint64
59         +---ro rx-passed-pkts?      uint64

```

```

1      +---ro rx-discarded-pkts?          uint64
2      +---ro rx-lost-pkts?              uint64
3      +---ro rx-tagless-pkts?          uint64
4      +---ro rx-resets?                 uint64
5      +---ro rx-latent-error-resets?    uint64
6      +---ro encode-errored-pkts?      uint64

```

7

8 12.6.2 YANG data module definitions

9 12.6.2.4 Definition for the ieee802-dot1cb-frer YANG module

10 *Change the text as shown:*

```

11 module ieee802-dot1cb-frer {
12   yang-version "1.1";
13   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1cb-frer;
14   prefix dot1cb-frer;
15   import ieee802-dot1cb-stream-identification {
16     prefix dot1cb-sid;
17   }
18   import ieee802-dot1cb-stream-identification-types {
19     prefix dot1cb-sid-types;
20   }
21   import ieee802-dot1cb-frer-types {
22     prefix dot1cb-frer-types;
23   }
24   import ieee802-dot1q-types {
25     prefix dot1qtypes;
26   }
27   import ietf-interfaces {
28     prefix if;
29   }
30   organization
31     "Institute of Electrical and Electronics Engineers";
32   contact
33     "WG-URL: http://ieee802.org/1/
34     WG-EMail: stds-802-1-1@ieee.org
35
36     Contact: IEEE 802.1 Working Group Chair
37     Postal: C/O IEEE 802.1 Working Group
38             IEEE Standards Association
39             445 Hoes Lane
40             Piscataway, NJ 08854
41             USA
42
43     E-mail: stds-802-1-chairs@ieee.org";
44   description
45     "Management objects that control the frame replication and
46     elimination from IEEE Std 802.1CB-2017. This YANG data model conforms
47     to the Network Management Datastore Architecture defined in RFC 8342.
48     Copyright (C) IEEE (2021). This version of this YANG module is part
49     of IEEE Std 802.1CBcv; see the standard itself for full legal
50     notices.";
51   revision 2024-08-22 {
52     description
53     "Published as part of IEEE Std 802.1CB-2017/Cor 1-2025.";

```

```

1 reference
2 "IEEE Std 802.1CB-2017/Cor 1-2025, Frame Replication and Elimination for
3 Reliability - Corrigendum 1.";
4 ]
5 revision 2021-12-08 {
6   description
7     "Published as part of IEEE Std 802.1CBcv-2021. Initial version.";
8   reference
9     "IEEE Std 802.1CBcv-2021, Frame Replication and Elimination for
10    Reliability - FRER YANG Data Model and Management Information Base
11    Module.";
12 }
13 feature auto-configuration {
14   description
15     "Autoconfiguration of entries in the tables for Stream identity
16     table, sequence recovery table, and sequence identification table.";
17   reference
18     "7.11 of IEEE Std 802.1CB-2017
19     10.7 of IEEE Std 802.1CB-2017";
20 }
21 typedef lan-path-id {
22   type int8;
23   description
24     "An integer specifying a path or LAN. If and only if a packet
25     matches an entry in the Sequence identification table that
26     specifies HSR or PRP in its frerSeqEncEncapsType object,
27     tsnStreamIdLanPathId specifies the LanId or PathId value that must
28     be matched for this tsnStreamIdEntry to apply. A value of -1
29     indicates that the LanId or PathId are to be ignored.";
30   reference
31     "10.22 of IEEE Std 802.1CB-2017";
32 }
33 typedef sequence-history-length {
34   type uint32 {
35     range "2..max";
36   }
37   description
38     "An integer specifying how many bits of the SequenceHistory
39     variable are to be used.";
40   reference
41     "7.4.3.2.2 of IEEE Std 802.1CB-2017";
42 }
43 grouping sequence-encode-decode {
44   description
45     "The sequence-encode-decode grouping indicates the type of
46     encapsulation used for this instance of the Sequence encode/decode
47     function.";
48   reference
49     "10.5.1.5 of IEEE Std 802.1CB-2017";
50   choice encapsulation {
51     description
52       "This choice indicates the type of encapsulation used for this
53       instance of the Sequence encode/decode function. The
54       encapsulation includes an Organizationally Unique Identifier
55       (OUI) or Company ID (CID) to identify the organization defining
56       the Sequence encode/decode method.";
57     reference
58       "10.5.1.5 of IEEE Std 802.1CB-2017";
59     container r-tag {

```

```

1  presence "true";
2  description
3    "R-TAG";
4  reference
5    "7.8 of IEEE Std 802.1CB-2017";
6  leaf type-number {
7    type dot1cb-frer-types:seq-encaps-method;
8    default "r-tag";
9    config false;
10   description
11     "The type number used for the R-TAG encode/decode method.";
12   reference
13     "10.5.1.5 of IEEE Std 802.1CB-2017";
14 }
15 leaf oui-cid {
16   type string {
17     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
18   }
19   default "00-80-C2";
20   config false;
21   description
22     "The Organizationally Unique Identifier (OUI) or Company ID
23     (CID) to identify the organization defining the encode/decode
24     method. For encode/decode methods defined in IEEE Std
25     802.1CB-2017 the OUI/CID is always 00-80-C2.";
26   reference
27     "10.5.1.5 of IEEE Std 802.1CB-2017";
28 }
29 }
30 container hsr-sequence-tag {
31   presence "true";
32   description
33     "HSR sequence tag";
34   reference
35     "7.9 of IEEE Std 802.1CB-2017";
36   leaf type-number {
37     type dot1cb-frer-types:seq-encaps-method;
38     default "hsr-seq-tag";
39     config false;
40     description
41       "The type number used for the HSR sequence tag encode/decode
42       method.";
43     reference
44       "10.5.1.5 of IEEE Std 802.1CB-2017";
45   }
46   leaf oui-cid {
47     type string {
48       pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
49     }
50     default "00-80-C2";
51     config false;
52     description
53       "The Organizationally Unique Identifier (OUI) or Company ID
54       (CID) to identify the organization defining the encode/decode
55       method. For encode/decode methods defined in IEEE Std
56       802.1CB-2017 the OUI/CID is always 00-80-C2.";
57     reference
58       "10.5.1.5 of IEEE Std 802.1CB-2017";
59   }

```

```

1      }
2      container prp-sequence-tag {
3          presence "true";
4          description
5              "PRP sequence trailer";
6          reference
7              "7.10 of IEEE Std 802.1CB-2017";
8          leaf type-number {
9              type dot1cb-frer-types:seq-encaps-method;
10             default "prp-seq-trailer";
11             config false;
12             description
13                 "The type number used for the PRP sequence trailer
14                 encode/decode method.";
15             reference
16                 "10.5.1.5 of IEEE Std 802.1CB-2017";
17         }
18         leaf oui-cid {
19             type string {
20                 pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
21             }
22             default "00-80-C2";
23             config false;
24             description
25                 "The Organizationally Unique Identifier (OUI) or Company ID
26                 (CID) to identify the organization defining the encode/decode
27                 method. For encode/decode methods defined in IEEE Std
28                 802.1CB-2017 the OUI/CID is always 00-80-C2.";
29             reference
30                 "10.5.1.5 of IEEE Std 802.1CB-2017";
31         }
32     }
33     container organization-specific {
34         presence "true";
35         description
36             "This container allows to select Sequence encode/decode types
37             that are defined by entities outside of IEEE 802.1.";
38         reference
39             "10.5.1.5 of IEEE Std 802.1CB-2017";
40         leaf type-number {
41             type int32 {
42                 range "256..max";
43             }
44             description
45                 "The type number used for an encode/decode method defined by
46                 an entity owning the OUI or CID for this encapsulation type.";
47             reference
48                 "10.5.1.5 of IEEE Std 802.1CB-2017";
49         }
50         leaf oui-cid {
51             type string {
52                 pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
53             }
54             description
55                 "The Organizationally Unique Identifier (OUI) or Company ID
56                 (CID) to identify the organization defining the encode/decode
57                 method.";
58             reference
59                 "10.5.1.5 of IEEE Std 802.1CB-2017";

```

```

1      }
2    }
3  }
4 }
5 grouping sequence-recovery-algorithm {
6   description
7     "The sequence-recovery-algorithm grouping specifies which sequence
8     recovery algorithm is to be used for this instance of the Sequence
9     recovery function.";
10  reference
11    "10.4.1.5 of IEEE Std 802.1CB-2017";
12  choice algorithm {
13    description
14      "This choice indicates the sequence recovery algorithm used for
15      this instance of the Sequence recovery function. It includes an
16      Organizationally Unique Identifier (OUI) or Company ID (CID) to
17      identify the organization defining the sequence recovery
18      algorithm.";
19    reference
20      "10.4.1.5 of IEEE Std 802.1CB-2017";
21    container vector {
22      presence "true";
23      description
24        "Vector Recovery Algorithm.";
25      reference
26        "10.4.1.5 of IEEE Std 802.1CB-2017";
27      leaf type-number {
28        type dot1cb-frer-types:seq-rcvy-algorithm;
29        default "vector";
30        config false;
31        description
32          "The type number used for the VectorRecoveryAlgorithm.";
33        reference
34          "10.4.1.5 of IEEE Std 802.1CB-2017";
35      }
36      leaf oui-cid {
37        type string {
38          pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
39        }
40        default "00-80-C2";
41        config false;
42        description
43          "The Organizationally Unique Identifier (OUI) or Company ID
44          (CID) to identify the organization defining the sequence
45          recovery algorithm. For sequence recovery algorithms defined
46          in IEEE Std 802.1CB-2017 the OUI/CID is always 00-80-C2.";
47        reference
48          "10.4.1.5 of IEEE Std 802.1CB-2017";
49      }
50    }
51    container match {
52      presence "true";
53      description
54        "Match Recovery Algorithm.";
55      reference
56        "10.4.1.5 of IEEE Std 802.1CB-2017";
57      leaf type-number {
58        type dot1cb-frer-types:seq-rcvy-algorithm;
59        default "match";

```

```

1      config false;
2      description
3          "The type number used for the MatchRecoveryAlgorithm.";
4      reference
5          "10.4.1.5 of IEEE Std 802.1CB-2017";
6  }
7  leaf oui-cid {
8      type string {
9          pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
10     }
11     default "00-80-C2";
12     config false;
13     description
14         "The Organizationally Unique Identifier (OUI) or Company ID
15         (CID) to identify the organization defining the sequence
16         recovery algorithm. For sequence recovery algorithms defined
17         in IEEE Std 802.1CB-2017 the OUI/CID is always 00-80-C2.";
18     reference
19         "10.4.1.5 of IEEE Std 802.1CB-2017";
20 }
21 }
22 container organization-specific {
23     presence "true";
24     description
25         "This container allows to select sequence recovery algorithms
26         that are defined by entities outside of IEEE 802.1.";
27     reference
28         "10.4.1.5 of IEEE Std 802.1CB-2017";
29     leaf type-number {
30         type int32 {
31             range "256..max";
32         }
33         description
34             "The type number used for a sequence recovery algorithm
35             defined by an entity owning the OUI or CID for this algorithm
36             type.";
37         reference
38             "10.4.1.5 of IEEE Std 802.1CB-2017";
39     }
40     leaf oui-cid {
41         type string {
42             pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
43         }
44         description
45             "The Organizationally Unique Identifier (OUI) or Company ID
46             (CID) to identify the organization defining the sequence
47             recovery algorithm.";
48         reference
49             "10.4.1.5 of IEEE Std 802.1CB-2017";
50     }
51 }
52 }
53 }
54 container frer {
55     description
56         "The managed objects that control Stream identification are
57         described in Clause 9 of IEEE Std 802.1CB-2017. The managed objects
58         that control FRER are described in Clause 10 of IEEE Std
59         802.1CB-2017 as follows:

```

- 1
- 2 a) General requirements on the behavior of counters are in 10.1.
- 3 b) The various tables of managed objects that can manage, in
- 4 detail, each individual Stream, are described in five
- 5 subclauses, including:
- 6 1) Additions to the Stream identity table required for
- 7 Autoconfiguration.
- 8 2) The Sequence generation table that configures instances of
- 9 the Sequence generation function;
- 10 3) The Sequence recovery table that configures instances of the
- 11 Individual recovery function, the Sequence recovery function,
- 12 and the Latent error detection function;
- 13 4) The Sequence identification table that configures instances
- 14 of the Sequence encode/decode function; and
- 15 5) The Stream split table that configures instances of the
- 16 Stream splitting function.
- 17 c) The managed objects that support the automatic configuration,
- 18 upon receipt of a packet, of entries in the first four of the
- 19 preceding tables, are described in the subclause on
- 20 Autoconfiguration.
- 21 d) The per-port, per-Stream packet counters that are kept by FRER
- 22 functions for inspection by network management entities are
- 23 described in 10.8, and the per-port (totaled over all Streams)
- 24 counters in 10.9.
- 25

The managed objects in the subclauses under 9.1 make it possible to configure more than one encapsulation for the same stream_handle subparameter on the same port. Similarly, the managed objects in the subclauses under 10.3 and 10.4 make it possible to configure more than one Sequence encode/decode function or more than one Sequence generation function for the same stream_handle subparameter. [The same value of stream_handle can be in the frerSeqGenStreamList of more than one frerSeqGenEntry or in the frerSeqRcvyStreamList of more than one frerSeqRcvyEntry.] A system shall return an error if an attempt is made to configure conflicting requirements upon that system.";

reference

"Clause 10 of IEEE Std 802.1CB-2017";

list sequence-generation {

key "index";

description

"There is one Sequence generation table in a system, and one entry in the Sequence generation table for each Sequence generation function. Each frerSeqGenEntry lists the Streams and direction for which a single instance of the Sequence generation function is to be placed.";

reference

"10.3 of IEEE Std 802.1CB-2017";

leaf index {

type uint32;

description

"Clause 10 of IEEE Std 802.1CB-2017 states that the same Stream handle can be present multiple times in the sequence generation table. Therefore this index leaf is being used to uniquely identify an entry in the sequence generation table.";

}

leaf-list stream {

type leafref {

path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';

```

1      }
2      min-elements 1;
3      description
4          "A list of stream_handle values, corresponding to the values of
5          the tsNStreamIdHandle objects in the Stream identity table, on
6          which this instance of the Sequence generation function is to
7          operate. The single instance of the Sequence generation
8          function created by this frerSeqGenEntry operates every packet
9          belonging to this Stream, regardless of the port on which it is
10         received.";
11     reference
12         "10.3.1.1 of IEEE Std 802.1CB-2017";
13 }
14 leaf direction-out-facing {
15     type dot1cb-sid-types:direction;
16     description
17         "An object indicating whether the Sequence generation function
18         is to be placed on the out-facing (True) or in-facing (False)
19         side of the port.";
20     reference
21         "10.3.1.2 of IEEE Std 802.1CB-2017";
22 }
23 leaf reset {
24     type boolean;
25     description
26         "A Boolean object indicating that the Sequence generation
27         function is to be reset by calling its corresponding
28         SequenceGenerationReset function. Writing the value True to
29         frerSeqGenReset triggers a reset; writing the value False has
30         no effect. When read, frerSeqGenReset always returns the value
31         False.";
32     reference
33         "10.3.1.3 of IEEE Std 802.1CB-2017";
34 }
35 }
36 list sequence-recovery {
37     key "index";
38     description
39         "There is one Sequence recovery table in a system, and one entry
40         in the Sequence recovery table for each Sequence recovery
41         function or Individual recovery function that can also be
42         present. The entry describes a set of managed objects for the
43         single instance of a Base recovery function and Latent error
44         detection function included in the Sequence recovery function or
45         Individual recovery function. Each frerSeqRcvyEntry lists the
46         Streams, ports, and direction for which instances of a Sequence
47         recovery function or Individual recovery function are to be
48         instantiated.";
49     reference
50         "10.4 of IEEE Std 802.1CB-2017";
51     leaf index {
52         type uint32;
53         description
54             "Clause 10 of IEEE Std 802.1CB-2017 states that the same Stream
55             handle can be present multiple times in the sequence recovery
56             table. Therefore this index leaf is being used to uniquely
57             identify an entry in the sequence recovery table.";
58     }
59     leaf-list stream {

```

```
1     type leafref {
2         path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
3     }
4     min-elements 1;
5     description
6         "A list of the stream_handle values, corresponding to the
7         values of the tsnStreamIdHandle objects in the Stream identity
8         table, to which the system is to apply the instance of the
9         Sequence recovery function or Individual recovery function.";
10    reference
11        "10.4.1.1 of IEEE Std 802.1CB-2017";
12    }
13    leaf-list port {
14        type if:interface-ref;
15        min-elements 1;
16        description
17            "The list of ports on each of which the system is to
18            instantiate the Sequence recovery function, or from which
19            received packets are to be fed to a single instance of the
20            Individual recovery function.";
21        reference
22            "10.4.1.2 of IEEE Std 802.1CB-2017";
23    }
24    leaf direction-out-facing {
25        type dot1cb-sid-types:direction;
26        description
27            "An object indicating whether the Sequence recovery function or
28            Individual recovery function is to be placed on the out-facing
29            (True) or in-facing (False) side of the port.";
30        reference
31            "10.4.1.3 of IEEE Std 802.1CB-2017";
32    }
33    leaf reset {
34        type boolean;
35        default "false";
36        description
37            "A Boolean object indicating that the Sequence recovery
38            function or Individual recovery function is to be reset by
39            calling its corresponding SequenceGenerationReset function.
40            Writing the value True to frerSeqRcvyReset triggers a reset;
41            writing the value False has no effect. When read,
42            frerSeqRcvyReset always returns the value False.";
43        reference
44            "10.4.1.4 of IEEE Std 802.1CB-2017";
45    }
46    container algorithm {
47        description
48            "This object is an enumerated value specifying which sequence
49            recovery algorithm is to be used for this instance of the
50            Sequence recovery function. The enumeration uses an OUI or CID
51            as shown in Table 10-1. The default value for
52            frerSeqRcvyAlgorithm is Vector_Alg (00-80-C2, 0).";
53        reference
54            "10.4.1.5 of IEEE Std 802.1CB-2017";
55        uses sequence-recovery-algorithm;
56    }
57    leaf history-length {
58        type sequence-history-length;
59        default "2";
```

```

1      description
2          "An integer specifying how many bits of the SequenceHistory
3          variable are to be used. The minimum and the default value is
4          2, maximum is the maximum allowed by the implementation. [Not
5          used if frerSeqRcvyAlgorithm = Match_Alg (00-80-C2, 1).]";
6      reference
7          "10.4.1.6 of IEEE Std 802.1CB-2017";
8  }
9  leaf reset-timeout {
10      type uint32;
11      units "ms";
12      description
13          "An unsigned integer specifying the timeout period in
14          milliseconds for the RECOVERY_TIMEOUT event.";
15      reference
16          "10.4.1.7 of IEEE Std 802.1CB-2017";
17  }
18  leaf invalid-sequence-value {
19      type uint32;
20      config false;
21      description
22          "A read-only unsigned integer value that cannot be encoded in a
23          packet as a value for the sequence_number subparameter, i.e.,
24          frerSeqRcvyInvalidSequenceValue is larger than or equal to
25          RecovSeqSpace.";
26      reference
27          "10.4.1.8 of IEEE Std 802.1CB-2017";
28  }
29  leaf take-no-sequence {
30      type boolean;
31      default "false";
32      description
33          "A Boolean value specifying whether packets with no
34          sequence_number subparameter are to be accepted (True) or not
35          (False). Default value False.";
36      reference
37          "10.4.1.9 of IEEE Std 802.1CB-2017";
38  }
39  leaf individual-recovery {
40      type boolean;
41      description
42          "A Boolean value specifying whether this entry describes a
43          Sequence recovery function or Individual recovery function.
44          a) True: The entry describes an Individual recovery function.
45             Packets discarded by the SequenceGenerationAlgorithm will
46             cause the variable RemainingTicks to be reset. There is no
47             Latent error detection function associated with this entry,
48             so frerSeqRcvyLatentErrorDetection cannot also be True.
49          b) False: The entry describes a Sequence recovery function.
50             Packets discarded by the SequenceGenerationAlgorithm will
51             not cause the variable RemainingTicks to be reset.";
52      reference
53          "10.4.1.10 of IEEE Std 802.1CB-2017";
54  }
55  leaf latent-error-detection {
56      type boolean;
57      description
58          "A Boolean value indicating whether an instance of the Latent
59          error detection function is to be instantiated along with the

```

```

1      Base recovery function in this Sequence recovery function or
2      Individual recovery function. frerSeqRcvyLatentErrorDetection
3      cannot be set True if frerSeqRcvyIndividualRecovery is also
4      True; an Individual recovery function does not include a Latent
5      error detection function.";
6      reference
7      "10.4.1.11 of IEEE Std 802.1CB-2017";
8  }
9  container latent-error-detection-parameters {
10     description
11     "The objects in the following subclauses are present if and
12     only if frerSeqRcvyIndividualRecovery is False.";
13     reference
14     "10.4.1.12 of IEEE Std 802.1CB-2017";
15     leaf difference {
16         type uint32;
17         description
18         "An integer specifying the maximum difference between
19         frerCpsSeqRcvyDiscardedPackets, and the product of
20         frerCpsSeqRcvyPassedPackets and (frerSeqRcvyLatentErrorPaths
21         - 1) that is allowed. Any larger difference will trigger the
22         detection of a latent error by the LatentErrorTest function.";
23         reference
24         "10.4.1.12.1 of IEEE Std 802.1CB-2017";
25     }
26     leaf period {
27         type uint32;
28         units "ms";
29         default "2000";
30         description
31         "The integer number of milliseconds that are to elapse
32         between instances of running the LatentErrorTest function. An
33         implementation can have a minimum value for
34         frerSeqRcvyLatentErrorPeriod, below which it cannot be set,
35         but this minimum shall be no larger than 1000 ms (1 s).
36         Default value 2000 (2 s).";
37         reference
38         "10.4.1.12.2 of IEEE Std 802.1CB-2017";
39     }
40     leaf paths {
41         type uint16;
42         description
43         "The integer number of paths over which FRER is operating for
44         this instance of the Base recovery function and Latent error
45         detection function.";
46         reference
47         "10.4.1.12.3 of IEEE Std 802.1CB-2017";
48     }
49     leaf reset-period {
50         type uint32;
51         units "ms";
52         default "30000";
53         description
54         "The integer number of milliseconds that are to elapse
55         between instances of running the LatentErrorReset function.
56         An implementation can have a minimum value for
57         LatentErrorReset, below which it cannot be set, but this
58         minimum shall be no larger than 1000 ms (1 s). Default value
59         30000 (30 s).";

```

```

1         reference
2             "10.4.1.12.4 of IEEE Std 802.1CB-2017";
3     }
4 }
5 }
6 list sequence-identification {
7     key "port direction-out-facing";
8     description
9         "There is one Sequence identification table per system, and one
10        entry in the Sequence identification table for each port and
11        direction for which an instance of the Sequence encode/decode
12        function is to be created. Each entry in the Sequence
13        identification table specifies a port and direction on which an
14        instance of the Sequence encode/decode function is to be
15        instantiated for a list of Streams.";
16     reference
17         "10.5 of IEEE Std 802.1CB-2017";
18     leaf-list stream {
19         type leafref {
20             path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
21         }
22         min-elements 1;
23         description
24             "A list of stream_handles, corresponding to the values of the
25             tsnStreamIdHandle objects in the Stream identity table, for
26             which the system is to use the same encapsulation for the
27             Sequence encode/decode function.";
28         reference
29             "10.5.1.1 of IEEE Std 802.1CB-2017";
30     }
31     leaf port {
32         type if:interface-ref;
33         description
34             "The port on which the system is to place an instance of the
35             Sequence encode/decode function.";
36         reference
37             "10.5.1.2 of IEEE Std 802.1CB-2017";
38     }
39     leaf direction-out-facing {
40         type dot1cb-sid-types:direction;
41         description
42             "An object indicating whether the Sequence encode/decode
43             function is to be placed on the out-facing (True) or in-facing
44             (False) side of the port.";
45         reference
46             "10.5.1.3 of IEEE Std 802.1CB-2017";
47     }
48     leaf active {
49         type boolean;
50         description
51             "A Boolean value specifying whether this frerSeqEncEntry is
52             passive (False), and therefore is used only to decode (extract
53             information from) input packets passing up the protocol stack,
54             or active (True), and therefore is used both for recognizing
55             input packets and for encoding output packets being passed down
56             the protocol stack.";
57         reference
58             "10.5.1.4 of IEEE Std 802.1CB-2017";
59     }

```

```

1  container encapsulation {
2    description
3      "An enumerated value indicating the type of encapsulation used
4      for this instance of the Sequence encode/decode function. The
5      type includes an OUI or CID.";
6    reference
7      "10.5.1.5 of IEEE Std 802.1CB-2017";
8    uses sequence-encode-decode;
9  }
10 leaf path-id-lan-id {
11   type lan-path-id;
12   description
13     "A 4-bit integer value to be placed in the PathId field of an
14     HSR sequence tag or the LanId field of a PRP sequence trailer
15     added to an output packet. This managed object is used only if:
16     a) The HSR sequence tag or the PRP sequence trailer is
17        selected by the frerSeqEncEncapsType object; and
18     b) frerSeqEncActive is False (passive)";
19   reference
20     "10.5.1.6 of IEEE Std 802.1CB-2017";
21 }
22 }
23 list stream-split {
24   key "port direction out-facingindex";
25   description
26     "There is one Stream split table per system, with one
27     frerSplitEntry per Stream splitting function per set of
28     stream_handle values. Each entry in the Stream split table
29     specifies a port and direction on which an instance of the Stream
30     splitting function is to be instantiated, and the list of
31     stream_handles specifying its operation.";
32   reference
33     "10.6 of IEEE Std 802.1CB-2017";
34   leaf index {
35     type uint32;
36     description
37       "Entry in the Stream split table.";
38     reference
39       "10.6 of IEEE Std 802.1CB-2017";
40   }
41   leaf port {
42     type if:interface-ref;
43     description
44       "The port on which the system is to place an instance of the
45       Stream splitting function performing the stream_handle
46       translations specified by frerSplitInputIdList and
47       frerSplitOutputIdList is to be placed.";
48     reference
49       "10.6.1.1 of IEEE Std 802.1CB-2017";
50   }
51   leaf direction-out-facing {
52     type dot1cb-sid-types:direction;
53     description
54       "An object indicating whether the instance of the Stream
55       splitting function performing the stream_handle translations
56       specified by frerSplitInputIdList and frerSplitOutputIdList is
57       to be placed on the out-facing (True) or in-facing (False) side
58       of the port.";
59     reference

```

```

1      "10.6.1.2 of IEEE Std 802.1CB-2017";
2  }
3  leaf-list input-id {
4      type leafref {
5          path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
6      }
7      min-elements 1;
8      description
9          "A list of stream_handles (tsnStreamIdHandle values) that are
10         to be split.";
11      reference
12          "10.6.1.3 of IEEE Std 802.1CB-2017";
13  }
14  leaf-list output-id {
15      type leafref {
16          path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
17      }
18      min-elements 1;
19      description
20          "A list of stream_handles (tsnStreamIdHandle values) into which
21         the input packet is to be split, one copy per item in the
22         frerSplitOutputIdList.";
23      reference
24          "10.6.1.4 of IEEE Std 802.1CB-2017";
25  }
26 }
27 container autoconfiguration {
28     if-feature "auto-configuration";
29     description
30         "Container for autoconfiguration managed objects.";
31     reference
32         "10.7 of IEEE Std 802.1CB-2017";
33     list sequence {
34         key "index";
35         description
36             "There is one Sequence autoconfiguration table per system. It
37             contains any number of table entries. No two (or more) entries
38             in the Sequence autoconfiguration table can have the same
39             values for frerAutSeqSeqEncaps, frerAutSeqTagged, and
40             frerAutSeqVlan on any given port. Each frerAutSeqEntry object
41             relates to a single class of Streams, and specifies how entries
42             are created (and destroyed) in the Stream identity table, the
43             Sequence recovery table, and the Sequence identification table.";
44         reference
45             "10.7.1 of IEEE Std 802.1CB-2017";
46         leaf index {
47             type uint32;
48             description
49                 "Entry in the sequence list referencing to a single class of
50                 Streams.";
51         }
52         container sequence-encapsulation {
53             description
54                 "An enumerated value, specifying which Sequence encode/decode
55                 function, and therefore, which type sequence_number encoding,
56                 is to be recognized for the purposes of Autoconfiguration.";
57             reference
58                 "10.7.1.1.1 of IEEE Std 802.1CB-2017";
59             uses sequence-encode-decode;

```

```
1      }
2      leaf-list receive-port {
3          type if:interface-ref;
4          min-elements 1;
5          description
6              "The list of ports to which this frerAutSeqEntry applies, and
7              on which Stream identification functions, Sequence
8              encode/decode functions, and Individual recovery functions
9              are to be autocreated.";
10         reference
11             "10.7.1.1.2 of IEEE Std 802.1CB-2017";
12     }
13     leaf tagged {
14         type enumeration {
15             enum tagged {
16                 value 1;
17                 description
18                     "A frame must have a VLAN tag to be matched.";
19             }
20             enum priority {
21                 value 2;
22                 description
23                     "A frame must be untagged, or have a VLAN tag with a VLAN
24                     ID = 0 to be matched.";
25             }
26             enum all {
27                 value 3;
28                 description
29                     "A frame is matched whether tagged or not.";
30             }
31         }
32         description
33             "An enumerated value indicating whether packets to be matched
34             by this frerAutSeqEntry are permitted to have a VLAN tag.";
35         reference
36             "10.7.1.1.3 of IEEE Std 802.1CB-2017";
37     }
38     leaf-list vlan {
39         type dot1qtypes:vlanid;
40         description
41             "A list of vlan_identifiers for the packet to match. A null
42             list matches all vlan_identifiers.";
43         reference
44             "10.7.1.1.4 of IEEE Std 802.1CB-2017";
45     }
46     leaf-list recovery-port {
47         type if:interface-ref;
48         min-elements 1;
49         description
50             "The list of ports on which Sequence recovery functions are
51             to be autocreated by this frerAutSeqEntry.";
52         reference
53             "10.7.1.1.5 of IEEE Std 802.1CB-2017";
54     }
55     leaf destruction-interval {
56         type uint64;
57         units "ms";
58         default "86400000";
59         description
```

```
1      "An integer number of milliseconds after which an idle set of
2      functions created by this frerAutSeqEntry can be destroyed. A
3      value of 0 indicates that idle autoconfigured functions are
4      not to be destroyed. Default value is 86 400 000 decimal (one
5      day).";
6      reference
7      "10.7.1.1.6 of IEEE Std 802.1CB-2017";
8  }
9  leaf reset-interval {
10     type uint64;
11     units "ms";
12     description
13     "The value used to fill frerSeqRcvyResetMSec when
14     autoconfiguring entries in the Sequence recovery table.";
15     reference
16     "10.7.1.1.7 of IEEE Std 802.1CB-2017";
17 }
18 container algorithm {
19     description
20     "The value used to fill frerSeqRcvyAlgorithm when
21     autoconfiguring entries in the Sequence recovery table.";
22     reference
23     "10.7.1.1.8 of IEEE Std 802.1CB-2017";
24     uses sequence-recovery-algorithm;
25 }
26 leaf history-length {
27     type sequence-history-length;
28     default "2";
29     description
30     "The value used to fill frerSeqRcvyHistoryLength when
31     autoconfiguring entries in the Sequence recovery table.";
32     reference
33     "10.7.1.1.9 of IEEE Std 802.1CB-2017";
34 }
35 leaf create-individual {
36     type boolean;
37     description
38     "A Boolean value. If True, the receipt of a packet that
39     triggers the autoconfiguration of a new tsNStreamIdEntry also
40     triggers the instantiation of a frerSeqRcvyEntry for an
41     Individual recovery function.";
42     reference
43     "10.7.1.1.10 of IEEE Std 802.1CB-2017";
44 }
45 leaf create-recovery {
46     type boolean;
47     description
48     "A Boolean value. If True, the receipt of a packet that
49     triggers the autoconfiguration of a new tsNStreamIdEntry can
50     also trigger the instantiation of a frerSeqRcvyEntry for a
51     Sequence recovery function.";
52     reference
53     "10.7.1.1.11 of IEEE Std 802.1CB-2017";
54 }
55 leaf latent-error-detection {
56     type boolean;
57     description
58     "A Boolean value. If True, the autoconfiguration of a new
59     Sequence recovery function also creates an associated Latent
```

```

1      Error Detection function.";
2      reference
3      "10.7.1.1.12 of IEEE Std 802.1CB-2017";
4  }
5  leaf latent-error-difference {
6      type int32;
7      description
8      "The value used to fill frerSeqRcvyLatentErrorDifference when
9      autoconfiguring entries in the Sequence recovery table.";
10     reference
11     "10.7.1.1.13 of IEEE Std 802.1CB-2017";
12 }
13 leaf latent-error-period {
14     type uint32;
15     units "ms";
16     default "2000";
17     description
18     "The value used to fill frerSeqRcvyLatentErrorPeriod when
19     autoconfiguring entries in the Sequence recovery table.";
20     reference
21     "10.7.1.1.14 of IEEE Std 802.1CB-2017";
22 }
23 leaf latent-error-reset-period {
24     type uint32;
25     units "ms";
26     default "30000";
27     description
28     "The value used to fill frerSeqRcvyLatentResetPeriod when
29     autoconfiguring entries in the Sequence recovery table.";
30     reference
31     "10.7.1.1.15 of IEEE Std 802.1CB-2017";
32 }
33 }
34 list output {
35     key "index";
36     description
37     "There is one Output autoconfiguration table per system. It
38     contains any number of frerAutOutEntry objects, each relating
39     to a single class of Streams specifying how active entries are
40     created in the Sequence identification table. No two (or more)
41     entries in the Output autoconfiguration table can include the
42     same port in their frerAutSeqReceivePortList objects.";
43     reference
44     "10.7.2 of IEEE Std 802.1CB-2017";
45     leaf index {
46         type uint32;
47         description
48         "Entry in the output list referencing to a single class of
49         Streams.";
50     }
51     leaf-list port {
52         type if:interface-ref;
53         min-elements 1;
54         description
55         "The list of ports to which this frerAutOutEntry applies, and
56         on which active Sequence encode/decode functions are to be
57         autocreated.";
58         reference
59         "10.7.2.1.1 of IEEE Std 802.1CB-2017";

```

```

1      }
2      container encapsulation {
3          description
4              "An enumerated value, specifying which Sequence encode/decode
5              function, and therefore, which type sequence_number encoding,
6              is to be used for autoconfigured Streams on the ports in
7              frerAutSeqReceivePortList.";
8          reference
9              "10.7.2.1.2 of IEEE Std 802.1CB-2017";
10         uses sequence-encode-decode;
11     }
12     leaf lan-path-id {
13         type lan-path-id;
14         description
15             "An integer specifying a path or LAN. If and only if
16             frerAutOutEncaps specifies HSR or PRP frerAutOutLanPathId
17             specifies the LanId or PathId value to be inserted into the
18             HSR sequence tag or PRP sequence trailer of autoconfigured
19             packets transmitted on the ports in
20             frerAutSeqReceivePortList.";
21         reference
22             "10.7.2.1.3 of IEEE Std 802.1CB-2017";
23     }
24 }
25 }
26 }
27 augment "/dot1cb-sid:stream-identity" {
28     if-feature "auto-configuration";
29     description
30         "Two managed objects augment each tsnStreamIdEntry in the Stream
31         identity table when Managed objects for autoconfiguration is
32         implemented.";
33     reference
34         "10.2 of IEEE Std 802.1CB-2017";
35     leaf auto-configured {
36         type boolean;
37         config false;
38         description
39             "A read-only Boolean value, supplied by the system, specifying
40             whether this entry was created explicitly (False) or via the
41             Sequence autoconfiguration table.";
42         reference
43             "10.2.1 of IEEE Std 802.1CB-2017";
44     }
45     leaf lan-path-id {
46         type lan-path-id;
47         description
48             "An integer specifying a path or LAN. If and only if a packet
49             matches an entry in the Sequence identification table that
50             specifies HSR or PRP in its frerSeqEncEncapsType object,
51             tsnStreamIdLanPathId specifies the LanId or PathId value that
52             must be matched for this tsnStreamIdEntry to apply. A value of -1
53             indicates that the LanId or PathId are to be ignored.";
54         reference
55             "10.2.2 of IEEE Std 802.1CB-2017";
56     }
57 }
58 augment "/if:interfaces/if:interface/if:statistics" {
59     description

```

```

1      "The following counters are the counters for frame replication and
2      elimination for reliability. All counters are unsigned integers. If
3      used on links faster than 650 000 000 bits per second, they shall
4      be 64 bits in length to ensure against excessively short wrap
5      times.
6
7      A Stream identification component shall implement the first two
8      counters provided in the stream-identification YANG module per-port
9      counters, input-packets and output-packets; the remainder of the
10     counters in the frer YANG module module are optional for such a
11     system.";
12     reference
13     "10.8 of IEEE Std 802.1CB-2017
14     10.9 of IEEE Std 802.1CB-2017";
15     container frer {
16         description
17         "This container contains the per-port as well as the
18         per-port-per-stream counters for frame replication and
19         elimination for reliability.";
20         reference
21         "10.8 of IEEE Std 802.1CB-2017
22         10.9 of IEEE Std 802.1CB-2017";
23         container per-port-counters {
24             config false;
25             description
26             "Contains the per-port counters for frame replication and
27             elimination for reliability. The following counters are
28             instantiated for each port on which any of the Stream
29             identification function, Sequencing function, or Sequence
30             encode/decode function is configured. The counters are indexed
31             by port number.";
32             leaf rx-passed-pkts {
33                 type uint64;
34                 config false;
35                 description
36                 "The frerCpSeqRcvyPassedPackets counter is incremented once
37                 for each packet passed up the stack by the
38                 VectorRecoveryAlgorithm or MatchRecoveryAlgorithm. Its value
39                 equals the sum (modulo the size of the counters) of all of
40                 the frerCpsSeqRcvyPassedPackets counters on this same port.";
41                 reference
42                 "10.9.1 of IEEE Std 802.1CB-2017";
43             }
44             leaf rx-discarded-pkts {
45                 type uint64;
46                 config false;
47                 description
48                 "The frerCpSeqRcvyDiscardPackets counter is incremented once
49                 for each packet discarded due to a duplicate sequence number
50                 or for being a rogue packet by any VectorRecoveryAlgorithm or
51                 MatchRecoveryAlgorithm on this port. Its value equals the sum
52                 (modulo the size of the counters) of all of the
53                 frerCpsSeqRcvyRoguePackets and frerCpsSeqRcvyDiscardedPackets
54                 counters on this same port.";
55                 reference
56                 "10.9.2 of IEEE Std 802.1CB-2017";
57             }
58             leaf encode-errored-pkts {
59                 type uint64;

```

```

1      config false;
2      description
3          "The frerCpSeqEncErroredPackets counter is incremented once
4          each time the Sequence encode/decode function receives a
5          packet that it is unable to decode successfully. Its value
6          equals the sum (modulo the size of the counters) of all of
7          the frerCpsSeqEncErroredPackets counters on this same port.";
8      reference
9          "10.9.2 of IEEE Std 802.1CB-2017";
10     }
11 }
12 list per-port-per-stream-counters {
13     key "direction-out-facing handle";
14     config false;
15     description
16         "Contains the per-port-per-stream counters for frame
17         replication and elimination for reliability. The following
18         counters are instantiated for each port on which any of the
19         Stream identification function, Sequencing function, or
20         Sequence encode/decode function is configured. The counters are
21         indexed by port number, facing (in-facing or out-facing), and
22         stream_handle value.";
23     reference
24         "10.8 of IEEE Std 802.1CB-2017";
25     leaf direction-out-facing {
26         type dot1cb-sid-types:direction;
27         description
28             "An object indicating whether the counters apply to
29             out-facing (True) or in-facing (False).";
30     }
31     leaf handle {
32         type leafref {
33             path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
34         }
35         description
36             "The according tsNStreamIdHandle for the counters.";
37     }
38     leaf generation-reset {
39         type uint64;
40         config false;
41         description
42             "The frerCpsSeqGenResets counter is incremented each time the
43             SequenceGenerationReset function is called.";
44         reference
45             "10.8.2 of IEEE Std 802.1CB-2017";
46     }
47     leaf rx-out-of-order-pkts {
48         type uint64;
49         config false;
50         description
51             "The frerCpsSeqRcvyOutOfOrderPackets counter is incremented
52             once for each packet accepted out-of-order by the
53             VectorRecoveryAlgorithm or MatchRecoveryAlgorithm.
54             Out-of-order means that the packet's sequence number is not
55             one more than the previous packet received.";
56         reference
57             "10.8.3 of IEEE Std 802.1CB-2017";
58     }
59     leaf rx-rogue-pkts {

```

```

1      type uint64;
2      config false;
3      description
4          "The frerCpsSeqRcvyRoguePackets counter is incremented once
5           for each packet discarded by the VectorRecoveryAlgorithm
6           because its sequence_number subparameter is more than
7           frerSeqRcvyHistoryLength from RecovSeqNum.";
8      reference
9          "10.8.4 of IEEE Std 802.1CB-2017";
10     }
11     leaf rx-passed-pkts {
12         type uint64;
13         config false;
14         description
15             "The frerCpsSeqRcvyPassedPackets counter is incremented once
16              for each packet passed up the stack by the
17              VectorRecoveryAlgorithm or MatchRecoveryAlgorithm.";
18         reference
19             "10.8.5 of IEEE Std 802.1CB-2017";
20     }
21     leaf rx-discarded-pkts {
22         type uint64;
23         config false;
24         description
25             "The frerCpsSeqRcvyDiscardedPackets counter is incremented
26              once for each packet discarded due to a duplicate sequence
27              number by the VectorRecoveryAlgorithm or
28              MatchRecoveryAlgorithm.";
29         reference
30             "10.8.6 of IEEE Std 802.1CB-2017";
31     }
32     leaf rx-lost-pkts {
33         type uint64;
34         config false;
35         description
36             "The frerCpsSeqRcvyLostPackets counter is incremented once
37              for each packet lost by the VectorRecoveryAlgorithm. A packet
38              is counted as lost if its sequence number is not received on
39              any ingress port.
40
41              NOTE-If per-source sequence numbering is used,
42              frerCpsSeqRcvyLostPackets can count, as lost, packets that
43              were sent to another destination, but not lost.";
44         reference
45             "10.8.7 of IEEE Std 802.1CB-2017";
46     }
47     leaf rx-tagless-pkts {
48         type uint64;
49         config false;
50         description
51             "The frerCpsSeqRcvyTaglessPackets counter is incremented once
52              for each packet received by the VectorRecoveryAlgorithm that
53              has no sequence_number subparameter.";
54         reference
55             "10.8.8 of IEEE Std 802.1CB-2017";
56     }
57     leaf rx-resets {
58         type uint64;
59         config false;

```

```

1      description
2          "The frerCpsSeqRcvyResets counter is incremented once each
3              time the SequenceRecoveryReset function is called.";
4      reference
5          "10.8.9 of IEEE Std 802.1CB-2017";
6  }
7  leaf rx-latent-error-resets {
8      type uint64;
9      config false;
10     description
11         "The frerCpsSeqRcvyLatentErrorResets counter is incremented
12             once each time the LatentErrorReset function is called.";
13     reference
14         "10.8.10 of IEEE Std 802.1CB-2017";
15 }
16 leaf encode-errored-pkts {
17     type uint64;
18     config false;
19     description
20         "The frerCpsSeqEncErroredPackets counter is incremented once
21             each time the Sequence encode/decode function receives a
22             packet that it is unable to decode successfully.";
23     reference
24         "10.8.11 of IEEE Std 802.1CB-2017";
25 }
26 }
27 }
28 }
29 }
30
31
32

```

1 Annex C

2 (informative)

3 C.2 Example 2: Various stack positions

4 Illustrated in Figure C-4 is a simple network utilizing FRER for a single Compound Stream, in which a relay
5 system serves as a proxy for an end system that has no FRER capability (see item h in 7.1.1). In this figure,
6 the numbers (1, 2, 3) are points in item e in 7.1.1. “Split” indicates that the Compound Stream is split into
7 two streams by the ordinary bridge multicast mechanism, with the {VLAN, destination address} altered by
8 Stream identification functions on output. [No Stream splitting function (7.7) is used.] “Seq.” and “Rec.”
9 indicate whether the Sequencing function is acting on transmitted packets (sequence generation) or received
10 packets (sequence recovery).

1

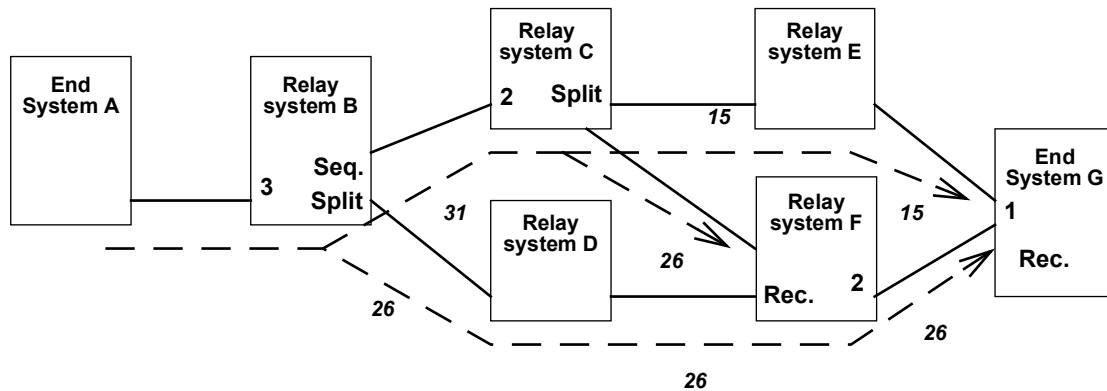


Figure C-4—Frame Replication and Elimination for Reliability RER flexible positioning

2 In Figure C-4, End System A is transmitting a Stream, but has no FRER functions. Relay system B
3 transforms that Stream into a Compound Stream by sequencing the packets and splitting them into Member
4 Streams 26 and 31 to go to relay systems C and D. Relay system C further splits the Stream into Streams 15
5 and 26 to go to relay systems E and F. (No sequencing is required; the packets were sequenced by relay
6 system B.) Relay system F combines the two Member Streams 26 from relay systems C and D, outputting a
7 single copy of each Packet on Stream 26 to End System G. End System G merges the two remaining
8 Member Streams 15 and 26, and discards the extras.

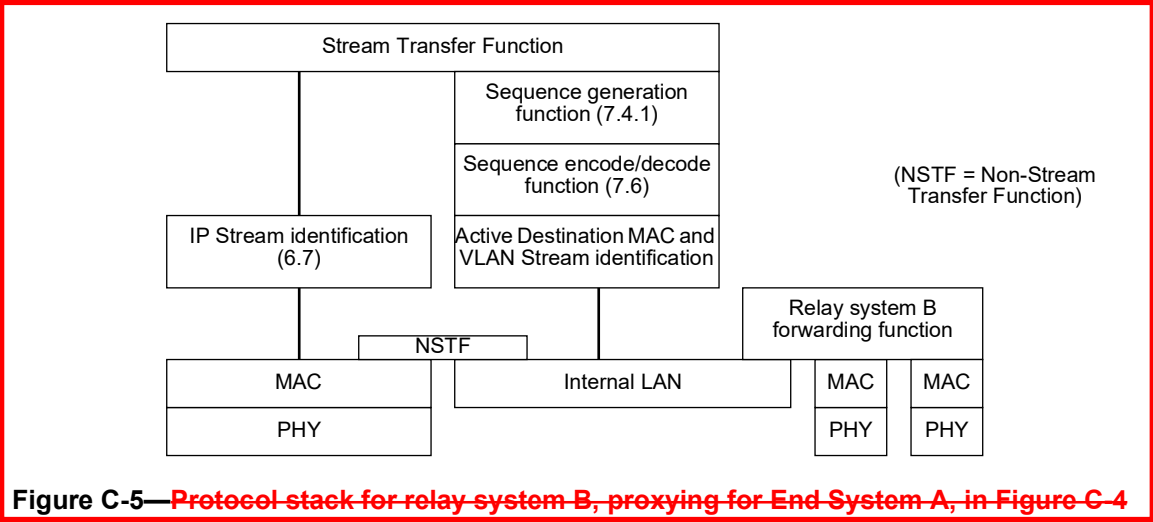
9 NOTE—In this example, stream_handle values are shown on the wires, being relayed from system to system. This is for
10 the convenience of the example. In an actual network, the stream_handle is an arbitrary integer that has meaning only
11 within a system, and is mapped to or from an external representation by the Stream identification functions (6.2) in the
12 various systems. That external representation can, but does not necessarily, include an explicit field corresponding to a
13 stream_handle subparameter.

14 Figure C-5 illustrates relay system B in Figure C-4. As the packets enter from the left, from End System A,
15 they pass first through a Stream identification function [IP Stream identification (6.7)], which identifies the
16 Stream. The Stream Transfer Function delivers the packet with all TSN parameters, including the
17 stream_handle subparameter, to the Sequence generation function (7.4.1, marked “Seq.” in Figure C-4),
18 which adds a sequence_number subparameter with a steadily-increasing integer sequence value (modulo the
19 size of the packet field carrying the sequence_number). The sequence_number subparameter is encapsulated
20 into the packet by the Sequence encode/decode function (7.6). A Stream identification function [this time,
21 In-facing Active Destination MAC and VLAN Stream identification (6.6)] modifies the two packets’
22 destination MAC addresses and VLANs for identification through the bridged network. Relay system B’s
23 forwarding function then outputs the two packets on two different ports. The external form of the packets are
24 labeled differently, as indicated by the italic numbers 26 and 31 in Figure C-4.

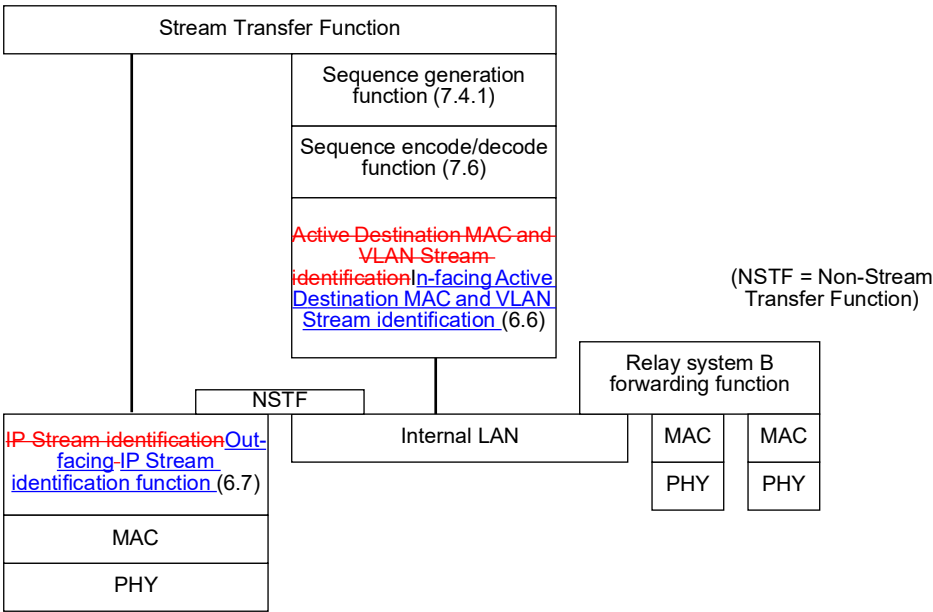
25 Figure C-6 illustrates relay system C in Figure C-4. The packets on Stream 31 enter from the left, from relay
26 system B. On output, they pass first through a passive Stream identification (Clause 6) function, which
27 identifies the Stream. The Stream Transfer Function delivers the packet with all TSN parameters, including
28 the stream_handle subparameter, to an active Stream identification (Clause 6) function. No Sequencing
29 function, Sequence generation function, or Sequence recovery functions are needed, because the packets
30 have already been sequenced (by relay system B) and none are being discarded. Stream identification
31 encapsulates the two packets differently on the two output ports (only one output port is shown in Figure C-6),
32 marking them as belonging to Streams 15 and 26, and relay system C’s forwarding function outputs the two
33 packets on two different ports.

34 **Change Figure C-5 as shown:**

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2



3

4

5

6

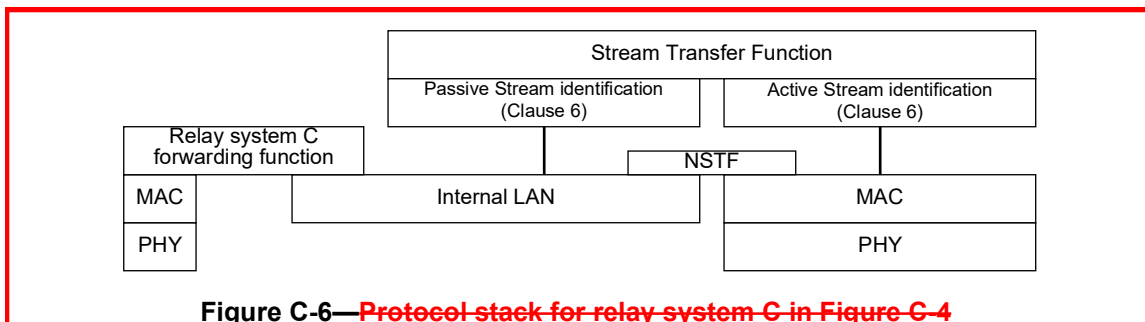
7

8

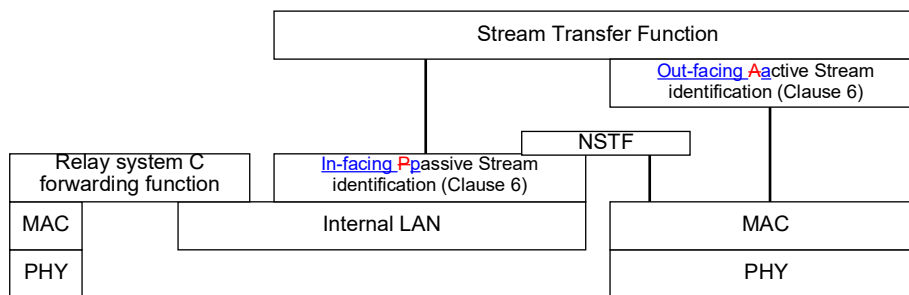
1

2 **Change Figure C-6 as shown:**

3



4



5 Figure C-7 illustrates relay system F in Figure C-4. In this case, the TSN layers are on the output port, rather
6 than the input port, as was the case for relay system B. Data packets from both Streams 26 are brought
7 together simply by the fact that they are both output to the same port. On that port, the In-facing passive
8 Stream identification function gives them both the same stream_handle subparameter, effectively merging
9 them into a single Stream. A Sequence encode/decode function (7.6) extracts the sequence_number
10 subparameters from the packets, so that the Sequence recovery function (7.4.2) can discard the replicates.
11 The Stream Transfer Function delivers all of these parameters to a Sequencing function and an Out-facing
12 active Stream identification function that re-encapsulate the packets, marking them all as belonging to
13 Stream 26, the same as when they were received. In this particular example, unlike Figure C-5, no
14 translation of Stream identification functions is being performed.

15 **Change Figure C-7 as shown:**

16 Finally, the protocol stack in the right-hand End System G of Figure C-4 is shown in Figure C-3. We assume
17 in this example that relay systems E and F use IEEE 802.1AX Distributed Resilient Network Interface
18 (DRNI), and that End System G uses IEEE 802.1AX Link Aggregation, in order to make End System G's
19 two physical ports appear, to the network, to be a single port, even though they are connected to two
20 different relay systems. The packets on Streams 15 and 26 are brought together by Link Aggregation. The
21 Stream identification function assigns the same stream_handle subparameter to packets of both Streams,
22 effectively merging them into a single Stream. They then pass through the Sequence recovery function,
23 which deletes the replicates for delivery to the upper layers. See Figure C-2 for a similar example of a
24 Talker, and C.6 for another type of two-port end system.

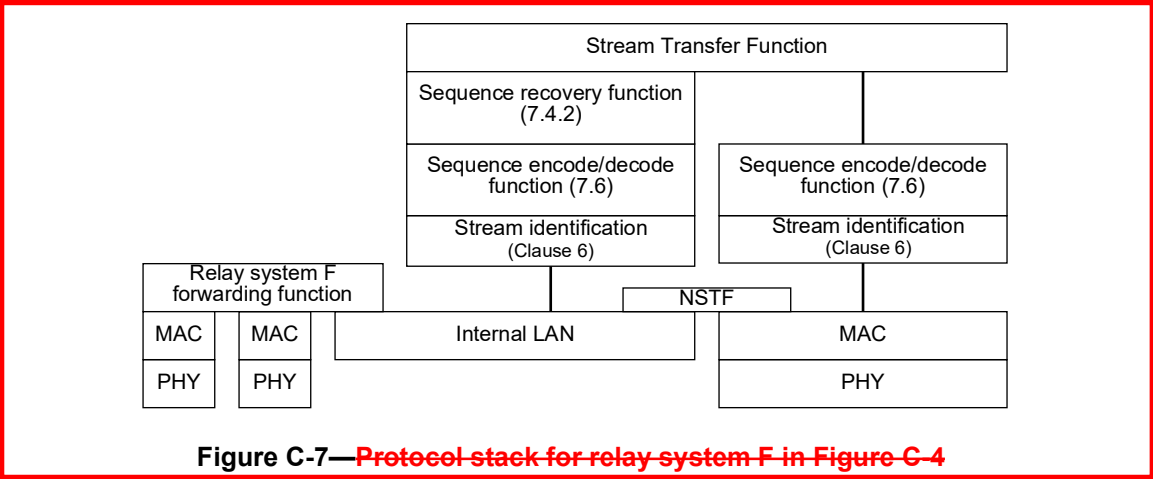


Figure C-7—Protocol stack for relay system F in Figure C-4

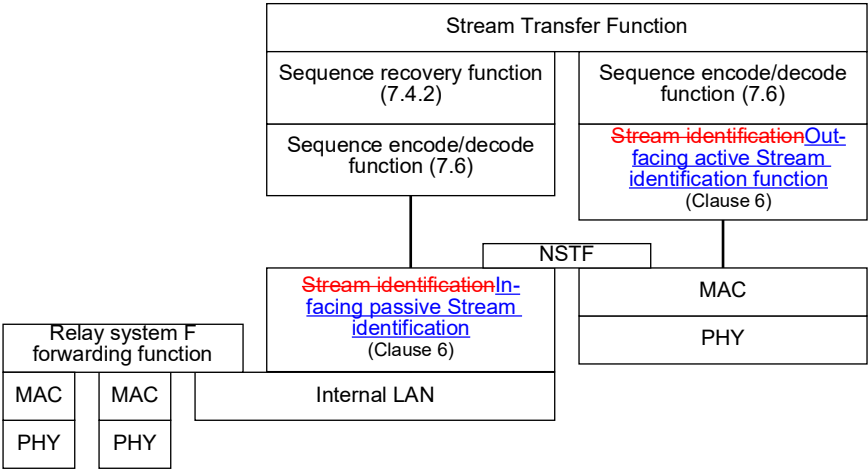


Figure C-7—Protocol stack for relay system F in Figure C-4

1

2 C.3 Example 3: Ladder redundancy

3 Illustrated in Figure C-8 is a network implementing “ladder redundancy.” This network will continue to
4 function in the face of multiple failures, because the Stream is repeatedly split and remerged.

5

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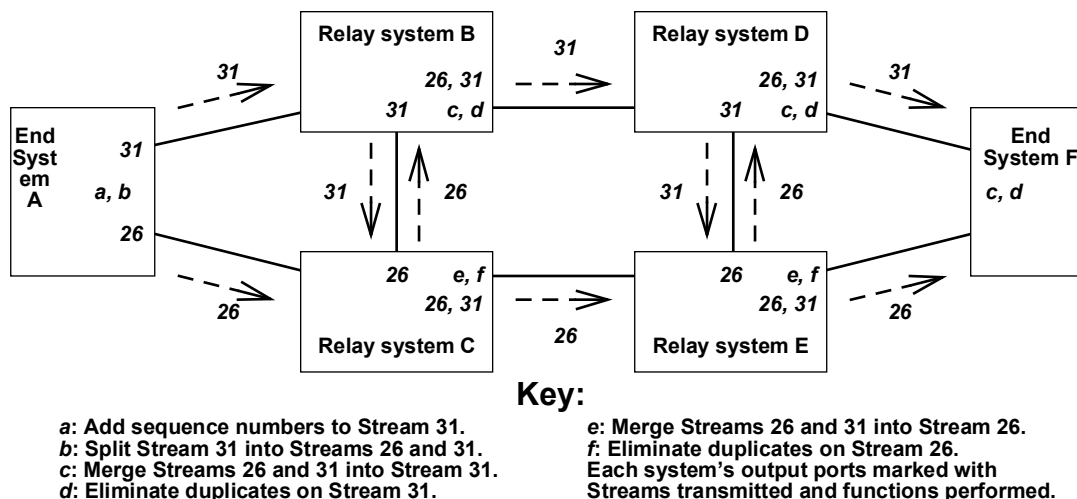


Figure C-8—Ladder redundancy

1 In Figure C-8, the Talker end system sequences Stream 31, then splits it into two Streams 26 and 31, sending
2 one on each of its two ports. The network “ladder” has two “rails,” an upper (relay systems B and D) and a
3 lower (relay systems C and E). The “rungs” are the connections B–C and D–E. Each relay system sends the
4 Stream received on the rail from the left both to the right, along the rail, and up or down, over the rung. It
5 forwards the packets received from the rung only to the right, along the rail. At each output port to the rail,
6 the end systems have a Sequence recovery function (7.4.2) to eliminate duplicates.

7 Each of the relay systems in Figure C-8 uses the protocol stack illustrated in Figure C-7.

8 C.5 Example 5: Protocol interworking

9 Figure C-10 illustrates a simple protocol interworking function in one port of a relay system. In this
10 example, two different encapsulation schemes **1** and **2** are used for the two legs of the Stream Transfer
11 Function, so that packets are transformed from using one encapsulation to using the other encapsulation as
12 they pass through the port. No additional functions, e.g., a Sequence recovery function (7.4.2) are shown,
13 although they would be perfectly admissible. If this were a port of a bridge attached to an end system,
14 encapsulation **1** could be the Active Destination MAC and VLAN Stream identification (6.6), and
15 encapsulation **2** could be the IP Stream identification (6.7). The net result for the end system could be to
16 convert a specific unicast IP Stream to use a specific multicast destination address and VLAN, in order to
17 direct the packet through a specific path through the bridged network. Presumably, a similar interworking
18 pair at the other end of the Stream would restore the packet to its original destination MAC address and
19 VLAN.

20 **Change Figure C-10 as shown:**

21

22

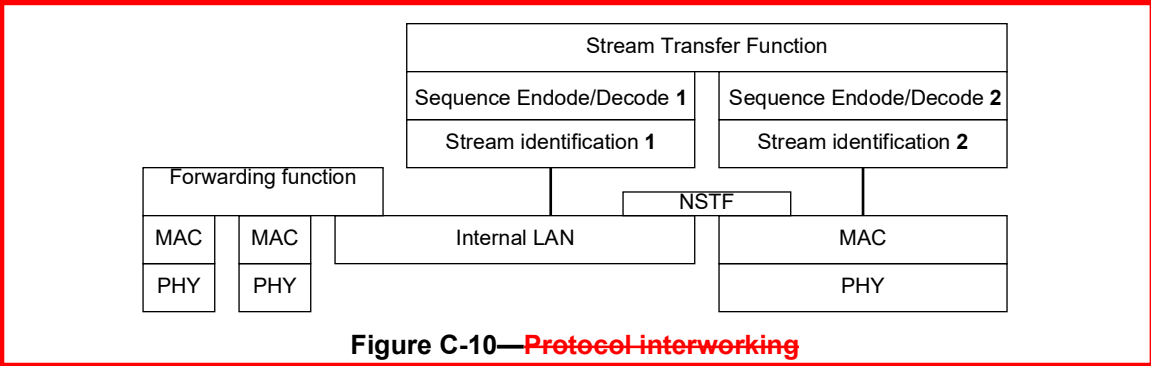


Figure C-10—Protocol interworking

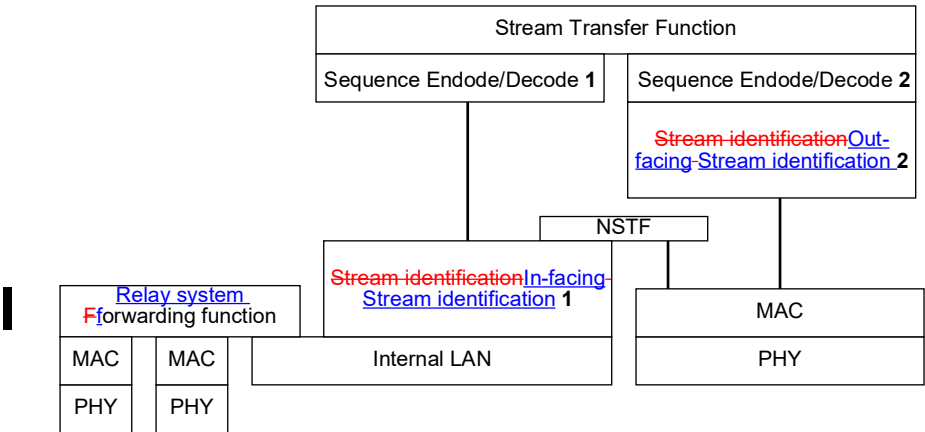


Figure C-10—Protocol interworking

1 C.6 Example 6: Chained two-port end systems

2 Illustrated in Figure C-11 is a simple network utilizing FRER for a single Compound Stream. In this
3 example, all FRER functions are confined to the two end systems B and G. “Seq.” and “Rec.” indicate
4 whether the Sequencing function is acting on transmitted packets (sequence generation) or received packets
5 (sequence recovery). In this example, unlike that in C.1, both end systems are composite systems; each uses
6 a separate one-port end system (B1, G1) attached to a three-port FRER C-component (B2, G2) in order to
7 connect their two ports (each) to a single protocol stack, as shown in Figure C-12.

8

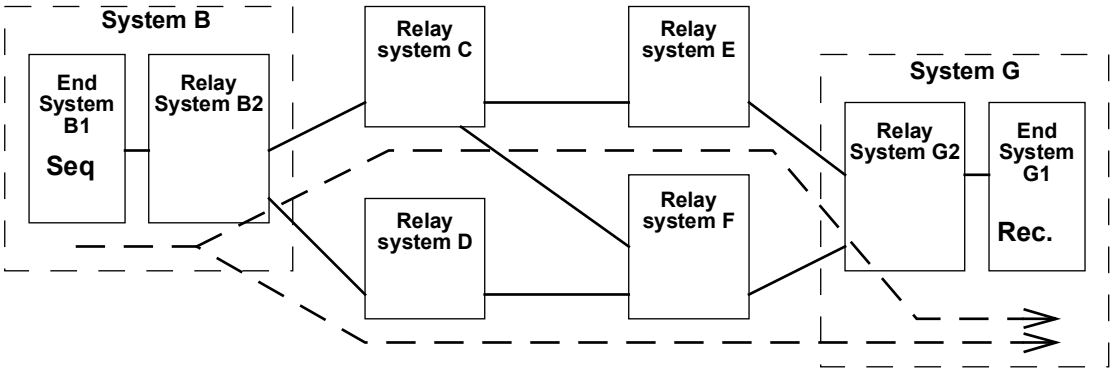
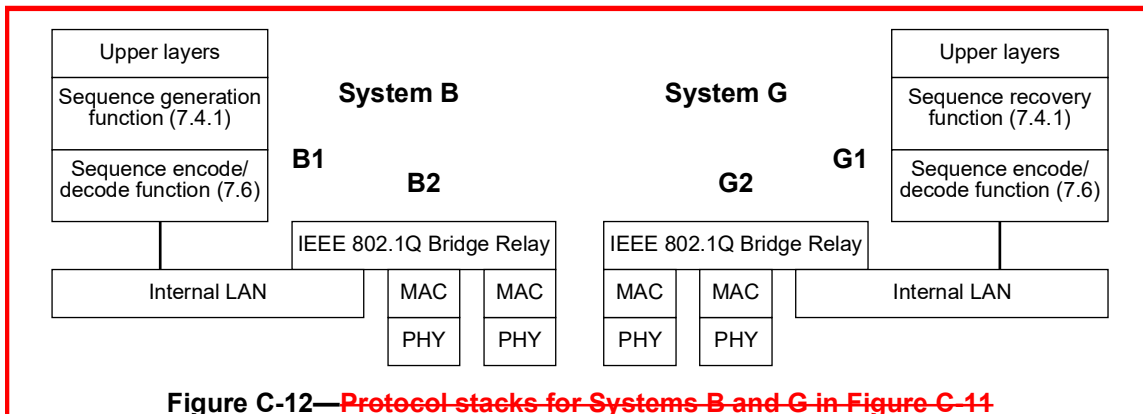
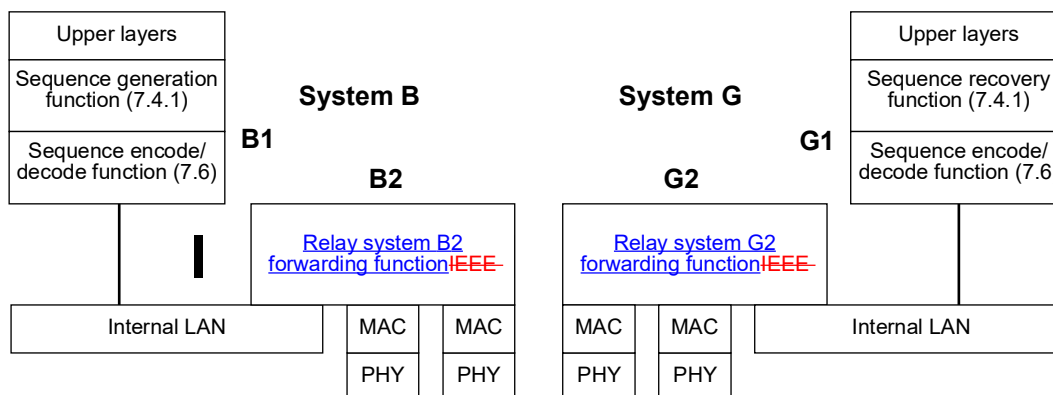


Figure C-11—Dual-homed end systems using 3-port bridge

1



2



3

4 In this example, we assume that the relay systems are all FRER C-components, and that the network has
5 been configured so that:

- 6 a) The Stream packets carry a multicast destination MAC address;
- 7 b) Bridge B2 has been configured to transmit the (multicast) Member Stream on both ports; and
- 8 c) The other Bridges, especially G2, have been configured to allow both Member Streams to enter
9 Bridge G2, but that Bridge G2 passes the Stream to end system G1 only (that is, the normal rules for
10 a multicast tree are broken).

11 Given this configuration, no Stream splitting function (7.7) is necessary for end system B1; packet
12 duplication is handled by the multicast forwarding functions performed in its associated relay system B2.

13 Unlike the example in C.1, the two-port composite end systems' Bridge functions (B2, G2) are required to
14 play their part in the network as Bridges, operating protocols and forwarding packets.